

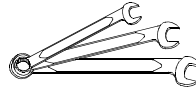
SAFETY PRECAUTIONS

☒	Stopped engine
☒	Block the starting mechanism
☒	Shut off starting air supply
☐	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☒	Shut off lubricating oil
☐	Lock turbocharger rotors

Data

Ref.	Description	Value	Unit
D-1	Weight of high-pressure pipe	34	kg
D-2	Tightening torque – High-pressure pipe	36	Nm

Standard Tools: See Section 913



10-120 Nm



Spare Parts

Plate – Item No.	Description	Qty
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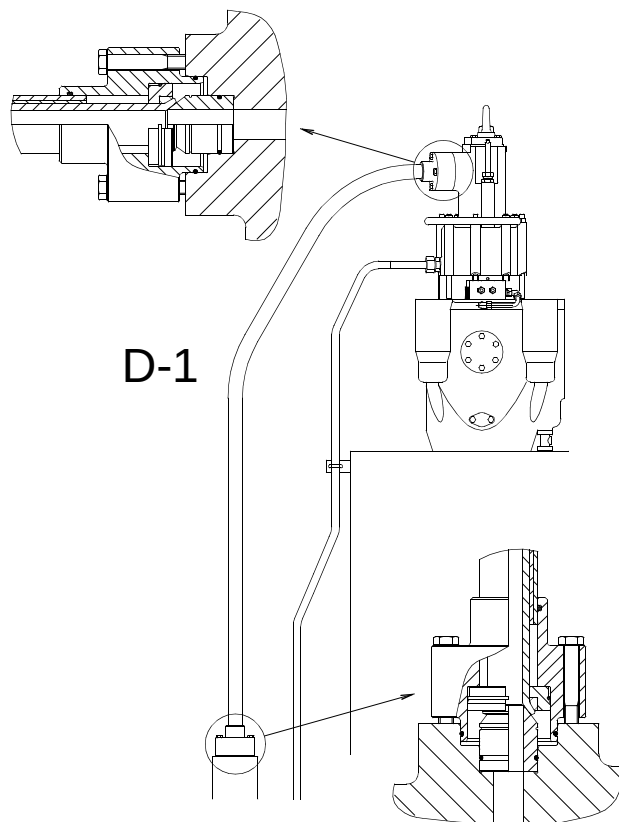
1. When replacing the hydraulic high-pressure pipe, exhaust valve or hydraulic actuator, check the high-pressure pipe sealing surfaces.

2. Before dismantling the high-pressure pipe, stop the camshaft oil pump.

Loosen the screws for the high-pressure pipe and lift the pipe slightly away from the actuator to let the oil in the pipe drain through the drain holes in the actuator.

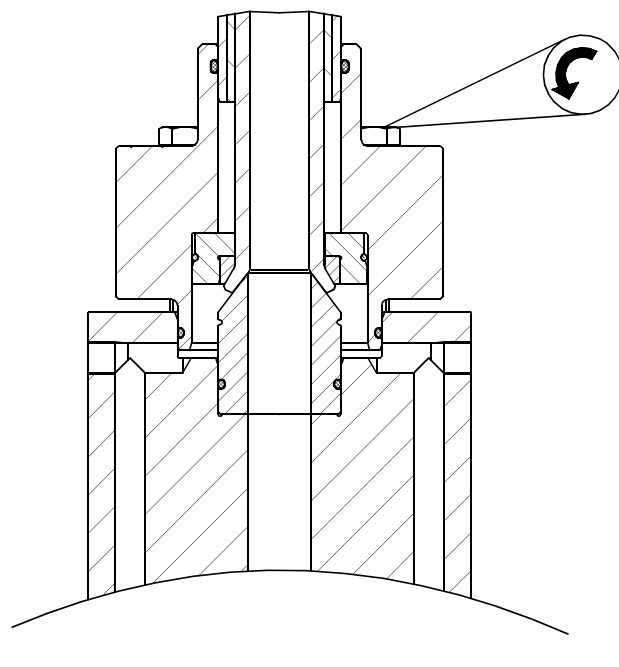
3. Remove the screws in both ends of the high-pressure pipe. Lift the high-pressure pipe away from the engine.

1.



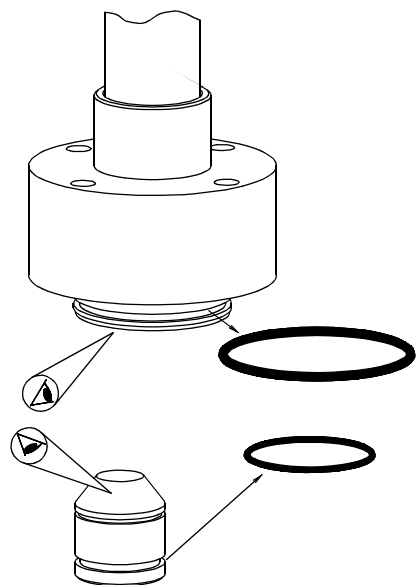
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2.



GN908-1.2 203 02

1.



The two ends of the pipe are identical and should both be overhauled as follows:

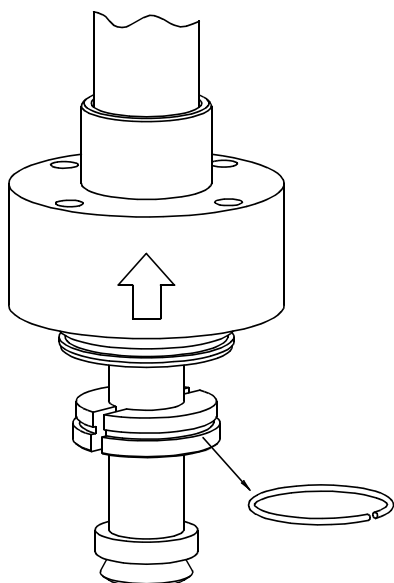
1. Remove the thrust piece from the actuator/exhaust valve.

Remove and discard the O-rings from the thrust piece and the thrust flange.

Clean and inspect the conical seats of the high-pressure pipe and the thrust piece.

Note!
The seats cannot be reconditioned. If a seat is damaged, the thrust piece/high-pressure pipe must be replaced.

2.



2. To change the internal O-ring, proceed as follows:

Lift the thrust flange upwards.

Remove the spring ring from the two-part sleeve.

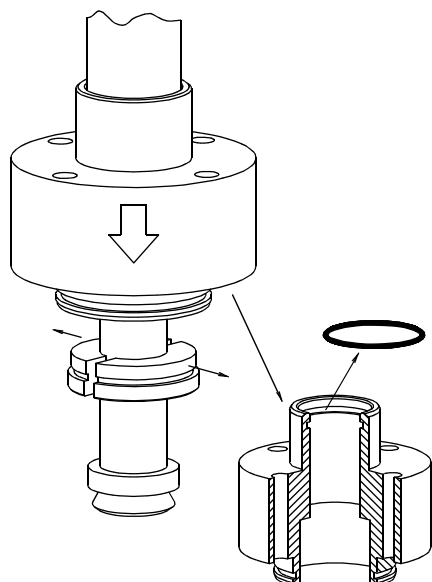
3. Remove the thrust flange from the pipe end.

Remove and discard the internal O-rings.

Mount new internal O-rings lubricated with a little camshaft lubricating oil.

4. Assemble the high-pressure pipe in the reverse order to disassembling.

3.



1. Mount new O-rings, lubricated with a little camshaft lubricating oil, on the thrust pieces and thrust flanges.

Mount the thrust pieces on the actuator/exhaust valve.

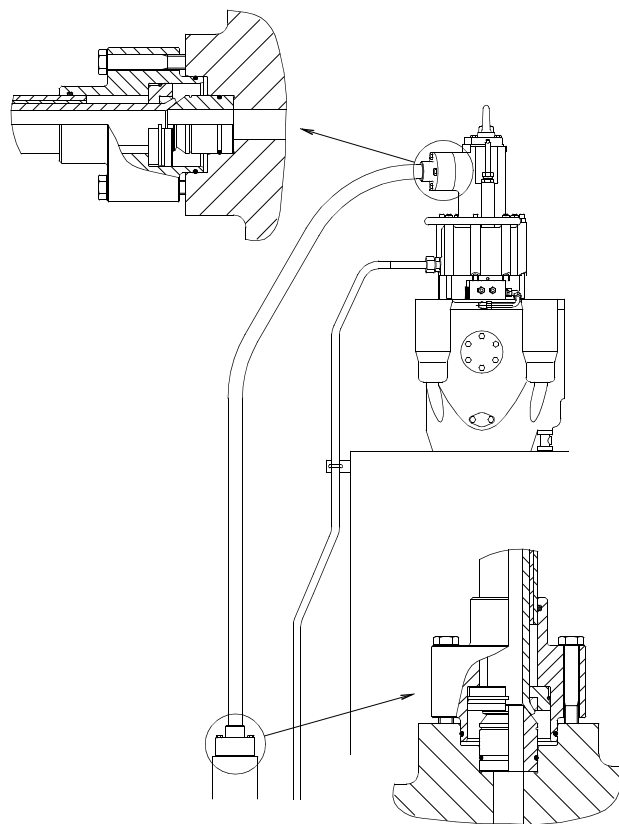
Mount the high-pressure pipe on the engine.

Note!

When mounting the high-pressure pipe, take care not to damage the conical seats of the pipe/thrust pieces.

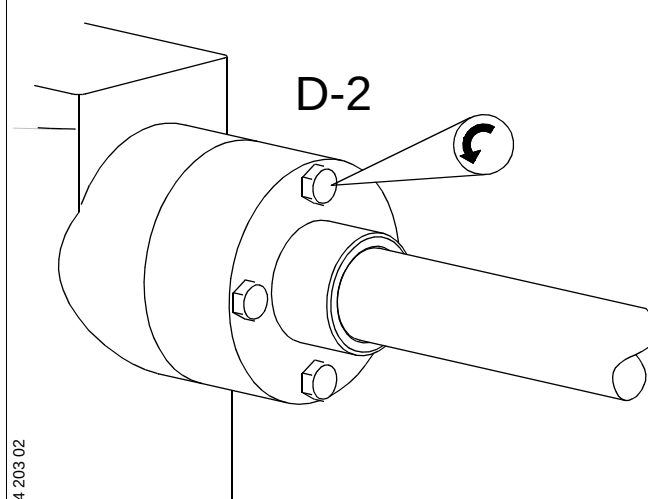
2. After fitting the pipe to the exhaust valve and actuator, tighten up the screws of the pressure flanges diagonally, see *Data*.
3. Check the system for tightness.

1.



GN908-1.2 203 01

2.



GN908-1.4 203 02

SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

X	Stopped engine
X	Shut off starting air supply – <i>At starting air receiver</i>
X	Block the main starting valve
X	Shut off starting air distributor/distributing system supply
X	Shut off safety air supply – <i>Not ME engines</i>
X	Shut off control air supply
X	Shut off air supply to exhaust valve – <i>Only with stopped lubricating oil pumps</i>
X	Engage turning gear
X	Shut off cooling water
	Shut off fuel oil
X	Stop lubricating oil supply
X	Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit
D08-04	Safety valve, opening pressure	23	bar
D08-05	Screws for lifting eye bolt, tightening torque	50	Nm
D08-06	Oil cylinder, tightening torque	70	Nm
D08-07	Safety valve, tightening torque	50 - 80	Nm
D08-09	Bottom piece seat, grinding angle	29.9 - 30.0	°
D08-10	Valve spindle seat, grinding angle	30.5 - 30.6	°
D08-11	Bottom piece seat, max. grinding	2.3	mm
D08-12	Gap of bottom piece seat	1.0	mm
D08-13	Spindle, max. burn-off	8.0	mm
D08-14	Spindle, max. grinding	2.0	mm
D08-15	Spindle stem, measuring area, min.	235	mm
D08-16	Spindle stem, measuring area, max.	345	mm
D08-17	Spindle stem, min. diameter	59.8	mm
D08-18	Bushing max. diameter, top	60.6	mm
D08-19	Bushing max. diameter, bottom	62.0	mm
D08-20	Oil cylinder max. inside diameter	71.2	mm
D08-21	Piston rings, min. thickness	2.6	mm
D08-24	Exhaust valve, complete	460	kg
D08-25	Exhaust valve housing	260	kg
D08-26	Oil cylinder	65	kg
D08-27	Air cylinder	20	kg
D08-28	Spindle	45	kg
D08-29	Bottom piece	55	kg
D08-30	Air piston	6	kg
D13-01	Hydraulic pressure, mounting	1500	bar
D13-02	Hydraulic pressure, dismantling	1400-1650	bar

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P90851	065	Lifting tool for exhaust valve
P90851	077	Cone ring for pneumatic piston
P90851	089	Pressure tester for safety valve
P90851	090	Gauge for exhaust valve spindle
P90851	150	Gauge for exhaust valve bottom piece
P90851	161	Lifting tool for exhaust valve spindle
P90851	185	Grinding ring for exhaust valve bottom piece
P90862		Exhaust Valve - Hydraulic Tools
P91351	010	Hydraulic pump, pneumatically operated
P91351	034	Hose with unions (1000 mm), complete
P91351	058	Hose with unions (3000 mm), complete
P91351	117	5-way distributor block, complete

Checks during running:**1. Check of valve spindle rotation**

The engine must run when the valve function and spindle rotation are checked. Lift the valve spindle rotation indicator, turn it 90° and let it rest on the air piston.

The indicator will now follow the movements of the valve spindle and the air piston. During rotation of the valve spindle, the top position of the indicator will change about 6 mm, because of the groove in the piston.

During engine operation, the spindle must rotate. The rotation can vary with unit and engine load. In the event of no rotation, attention should be paid to avoid seat problems. When check of the valve rotation is finished, lift the rotation indicator and lock it in the top position.

The indicator must **not** be in constant operation.

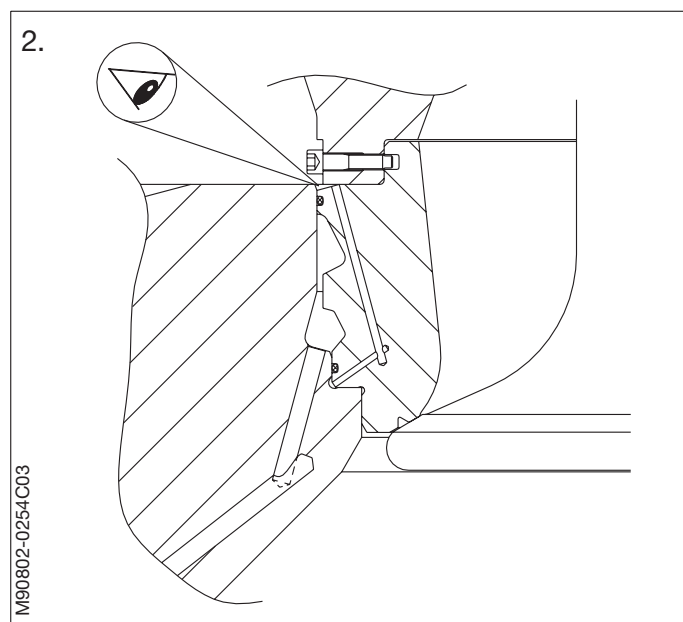
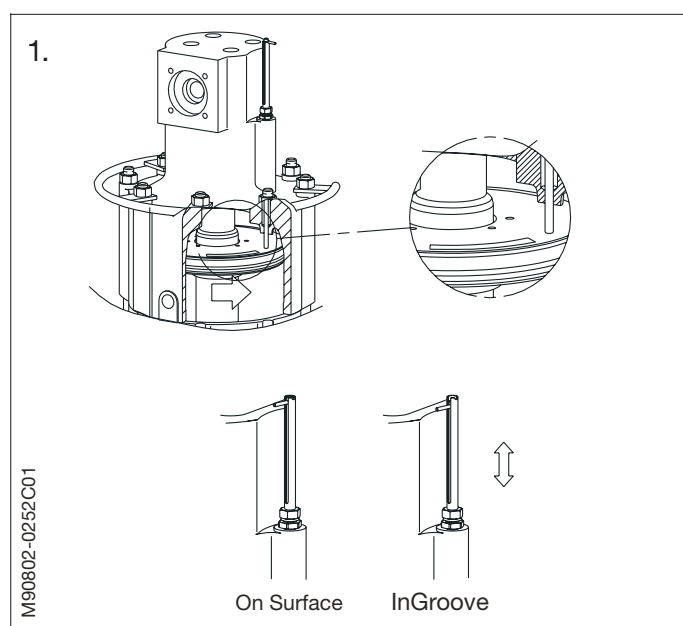
Exhaust valve performance condition
See volume I chapter 706 Performance Evaluation

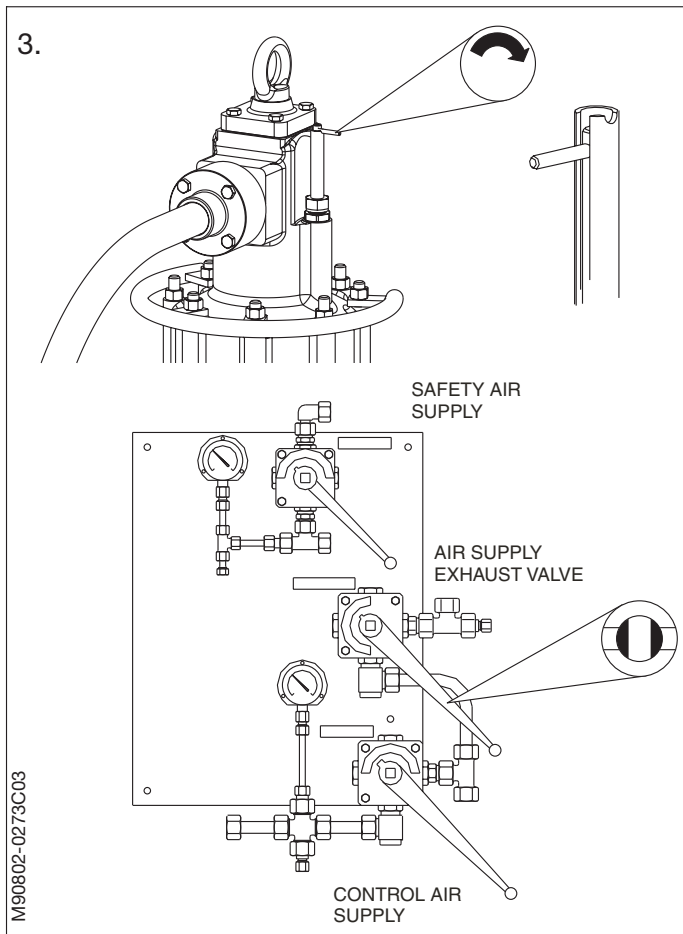
2. Check of bottom piece sealings

With the exhaust valve mounted in the cylinder cover, check the cooling water inlet and outlet to the cylinder cover and exhaust valve is open.

Check the tightness of the sealing rings between the bottom piece and the exhaust valve housing. IF water flows from the small bore on the manoeuvring side of the exhaust valve, the lowermost sealing ring is leaking.

If water flows from the any of the rings, the exhaust valve must be dismantled and the sealing rings replaced.



**Check with stopped engine:**

In connection with port stays, at an interval of e.g. one month, an easy check of all exhaust valves is recommended.

3. Check of spindle drop-down time:

Engage all rotation indicators on the exhaust valves.

Shut off and release the spring air on the valve, next to the engine-side manoeuvring stand.

Check the drop down time for all spindles:

- Spindles dropping down within 30 minutes: Check and overhaul, See next step "Check of Air Spring tightness".
- Spindles dropping down between 30 minutes and 1 hour: Keep under observation.
- Spindles dropping down after 1 hour: OK.

When check of drop down time is finished, lift the rotation indicators and lock them in the top position.

4. Check of Air spring tightness

Connect compressed air to the non return (A) valve on the air cylinder.

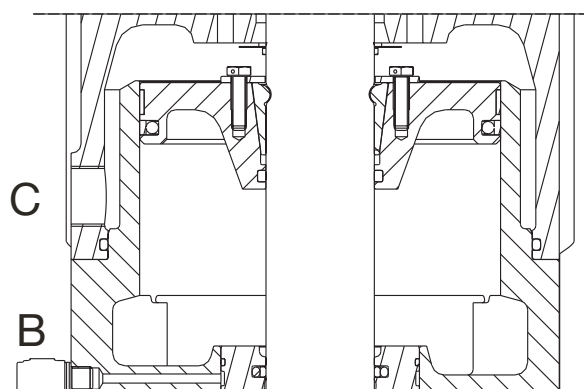
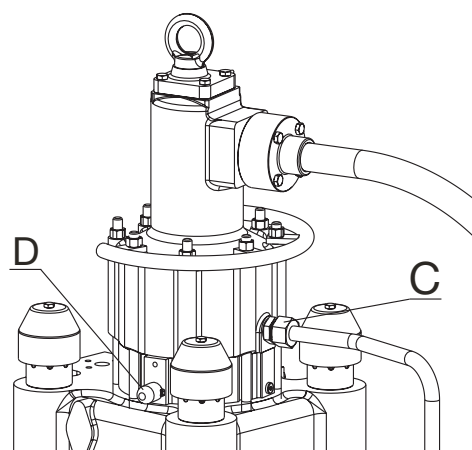
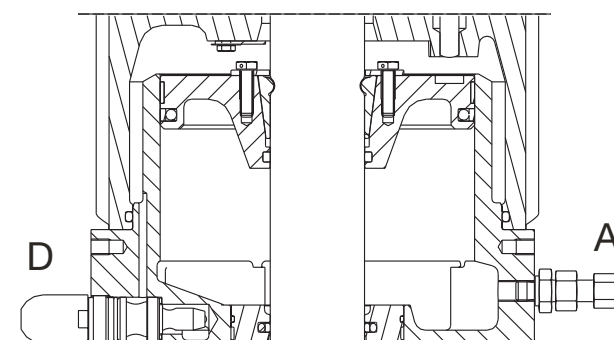
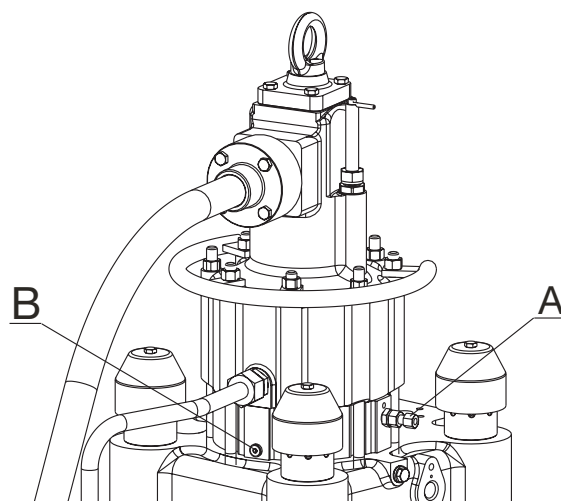
Disconnect and remove the drain oil pipe from the oil cylinder (C).

Check if air is coming out from:

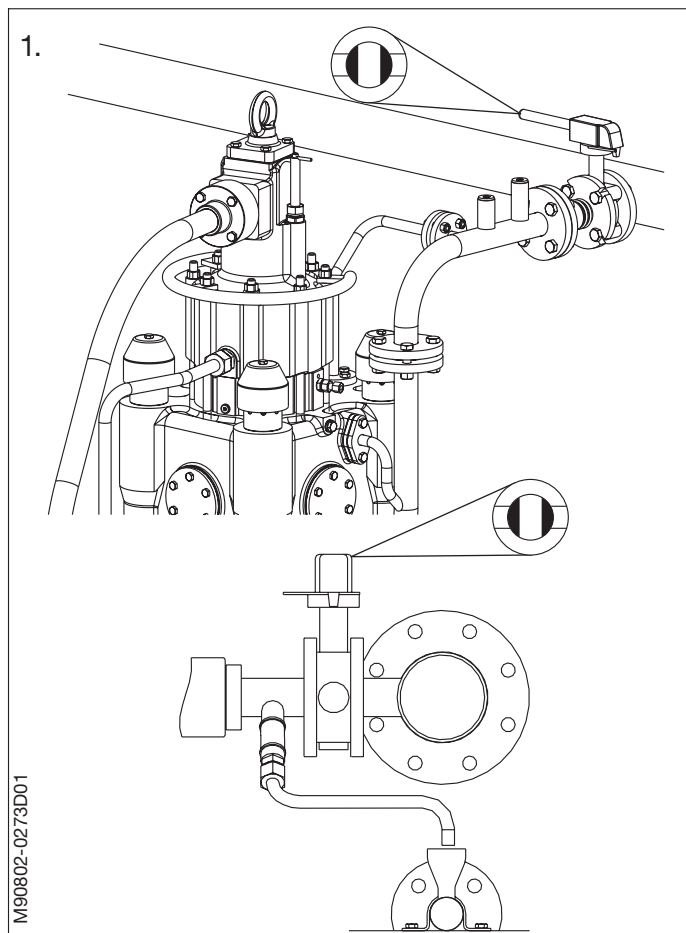
- The bore (B): Leakage from spindle stem sealing , change and overhaul.
- Hole for drain oil (C): If yes, the leakage comes from the safety valve or the air piston. Check the safety valve (D). If air still is coming out, the leakage comes from the air piston: change and overhaul.
- Remove the air supply to the non-return valve on the air cylinder and check the non-return valve for tightness.

The checks is for guidance only, minor leaks is difficult to detect.

4.



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**1. Close off cooling water**

Close the cooling water inlet and outlet connections, and drain the exhaust valve.

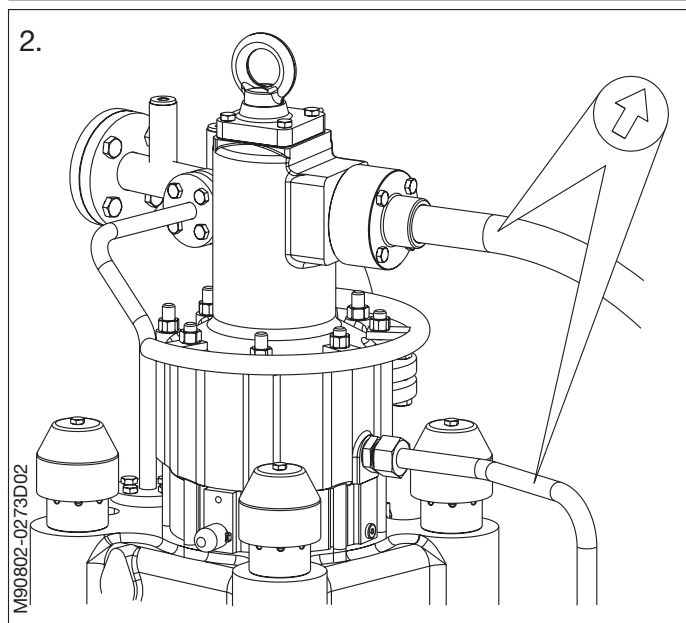
Close the air supply to the exhaust valve.

2. Dismount the oil pipe connections

Dismount the high-pressure pipe for the hydraulic valve actuation.

See *Procedure 908-1.2*

Dismount the return oil pipe from the exhaust valve.



3. Disconnect pipe connections

Disconnect the cooling water connections from the exhaust valve and the air pipe for the air spring.

Remove the plate jacket and insulation from the intermediate pipe.

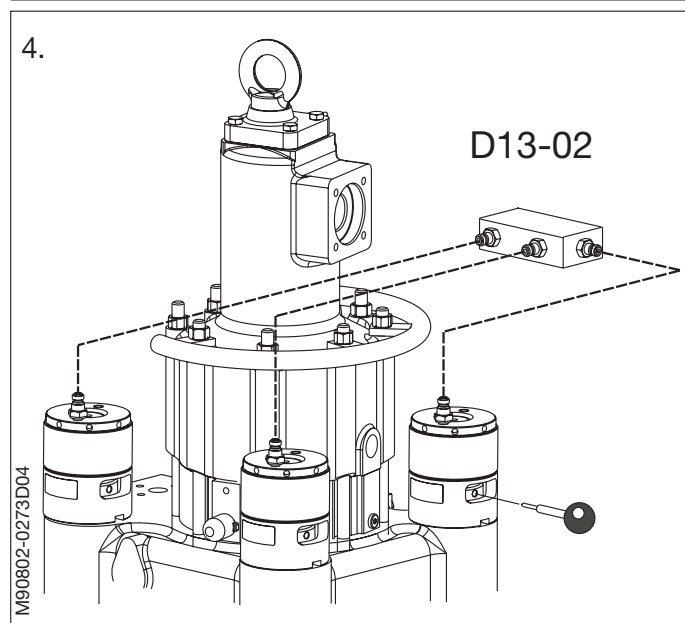
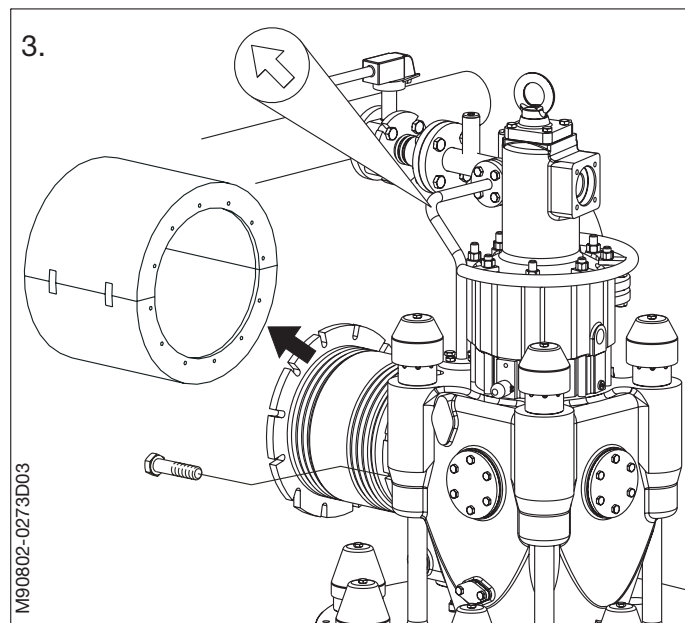
Loosen the screws which attach the intermediate pipe to the exhaust valve housing.

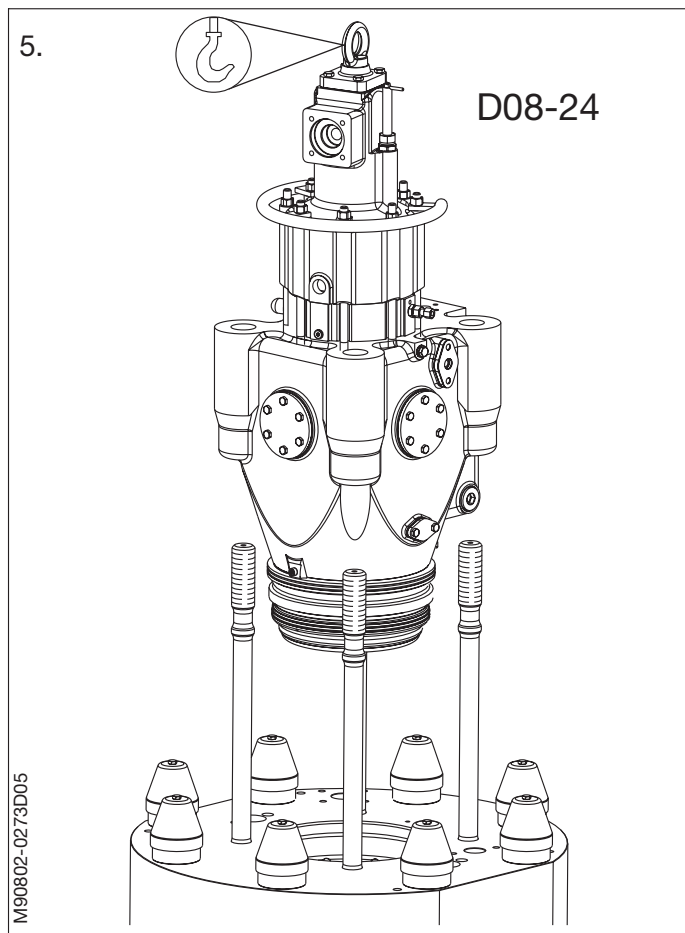
Support the intermediate pipe by e.g. a chain block and remove the screws.

4. Dismantling of hydraulic nuts.

Remove the protective caps from the exhaust valve studs.

Loosen and remove the hydraulic nuts, see Data. See *Procedure 913-1*.





5. Lift of exhaust valve

Check all connections to the valve are removed.

Attach the crane to the eye bolt fitted on top of the valve and start lifting the exhaust valve. During the lift - check the valve goes clear of all other engine components.

Land the exhaust valve on a wooden plate on the platform.

Carefully clean the exhaust valve bore in the cylinder cover and recondition the seating and sealing surfaces of the bore, if required.

See Procedure 901-1.3.

For overhaul of the exhaust valve, see Procedure 908-2.3

1. Oil cylinder

Place the exhaust valve on a wooden support on the platform.

Remove the nuts and the safety strap from the oil cylinder.

Lift away and place the oil cylinder on a wooden support.

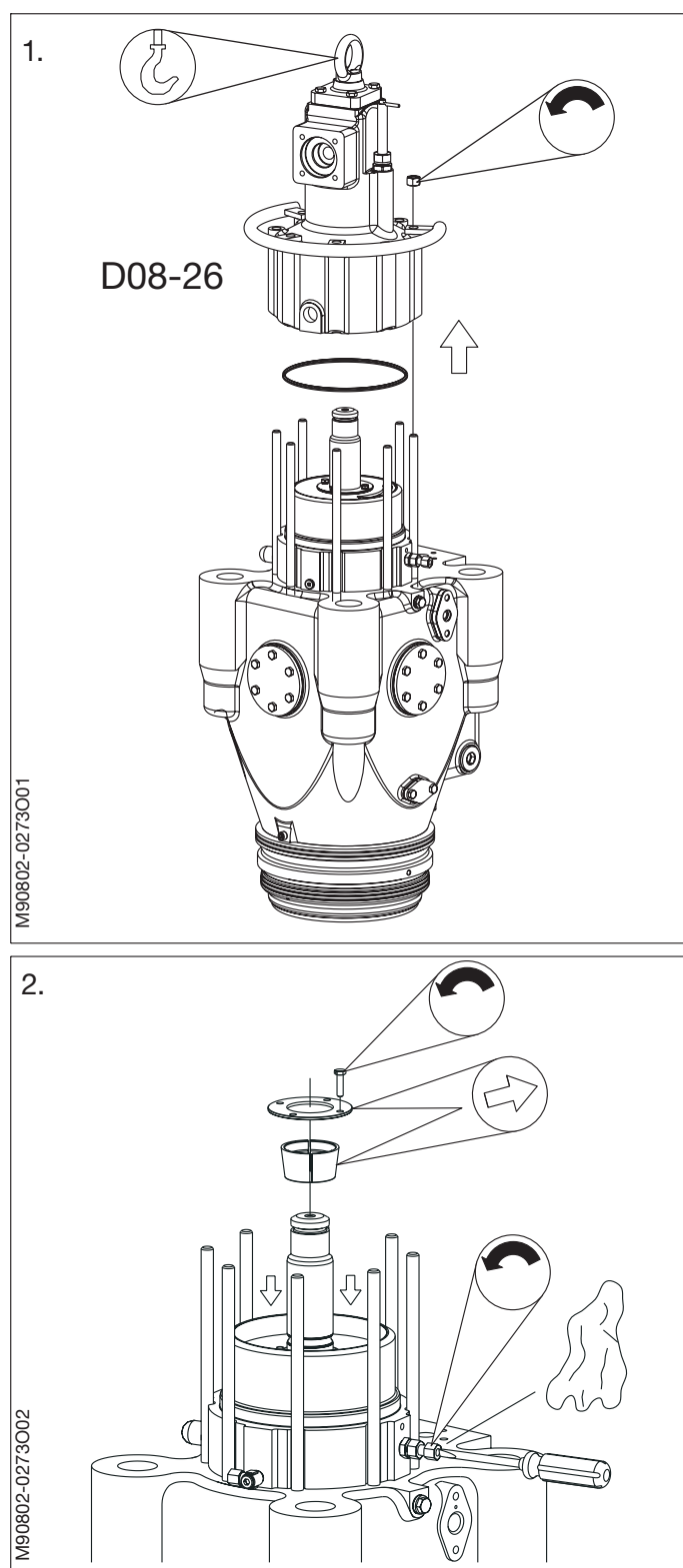
2. Conical locking ring

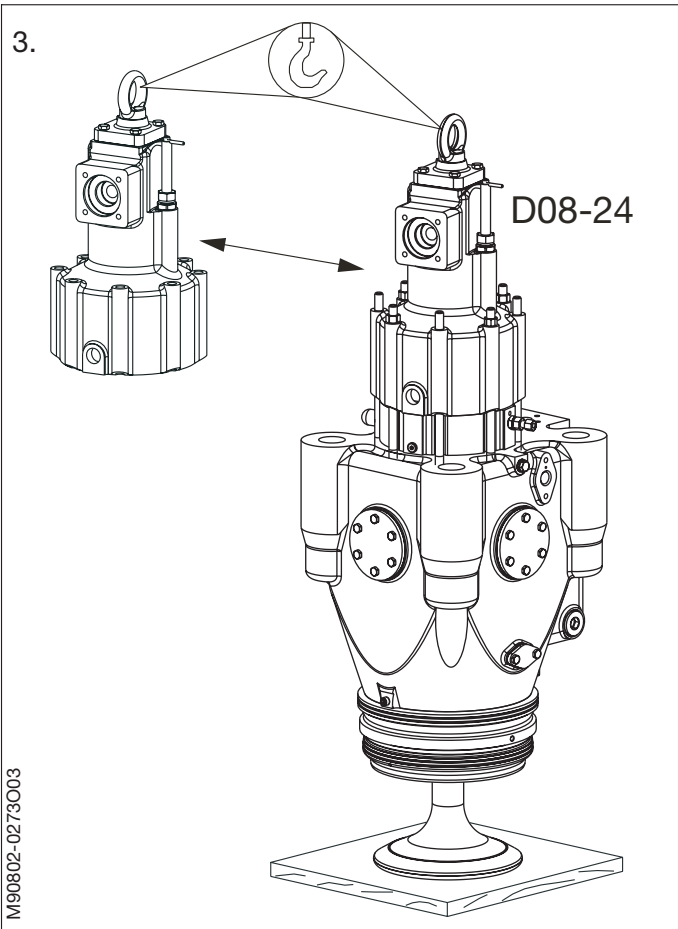
Relieve the air pressure through the non-return valve from below the piston. Use a small screwdriver to press the ball into the non-return valve. Cover the non-return valve with a rag to catch the oil drops.

Unscrew the non-return valve from the air cylinder.

Remove the four screws from the flange on top of the air piston and dismount the flange.

Loosen the air piston from the conical locking ring by means of a tin hammer, and remove the conical locking ring.





3. Spindle

Use the oil cylinder as a lifting tool for the exhaust valve housing. Lift the oil cylinder with the crane, and guide it down over the studs for the oil cylinder. Mount four nuts so that the load from the valve housing is evenly distributed.

Lift the valve housing clear of the valve spindle. **Take care** – when the internal O-ring in the air piston and the internal rings for the sealing arrangement at the bottom of the air cylinder pass the groove for the conical ring at the top of the spindle.

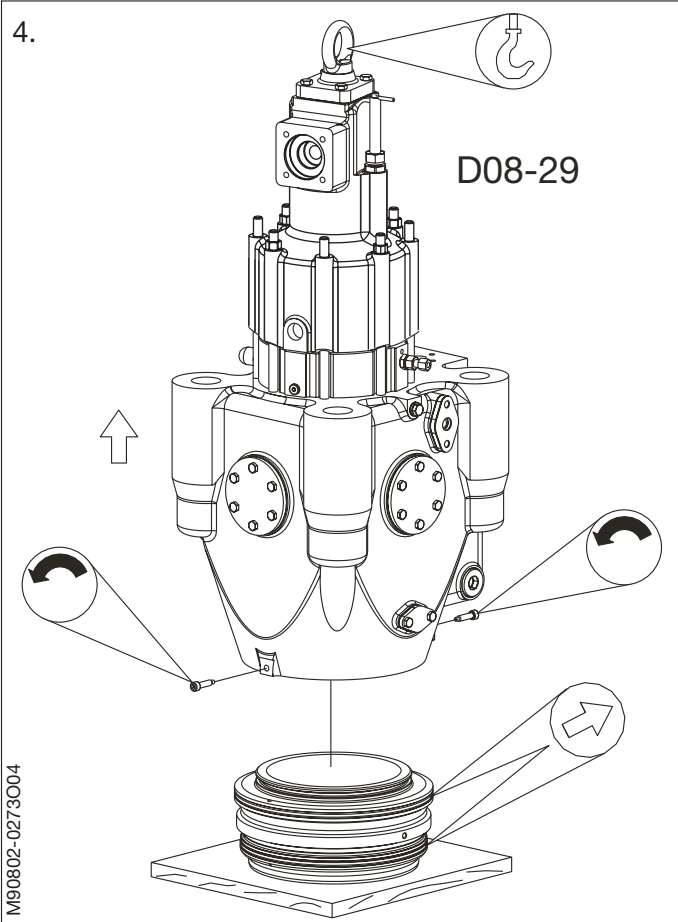
4. Bottom piece

Unscrew and remove the lock screws which retain the bottom piece. Lift the valve housing approx. 10 mm.

If the bottom piece has become stuck, use a tin hammer to release it.

Lift the valve housing away and land it on a couple of wooden planks.

Remove the oil cylinder.



5. Air piston and cylinder

Dismount the air piston from the air cylinder.

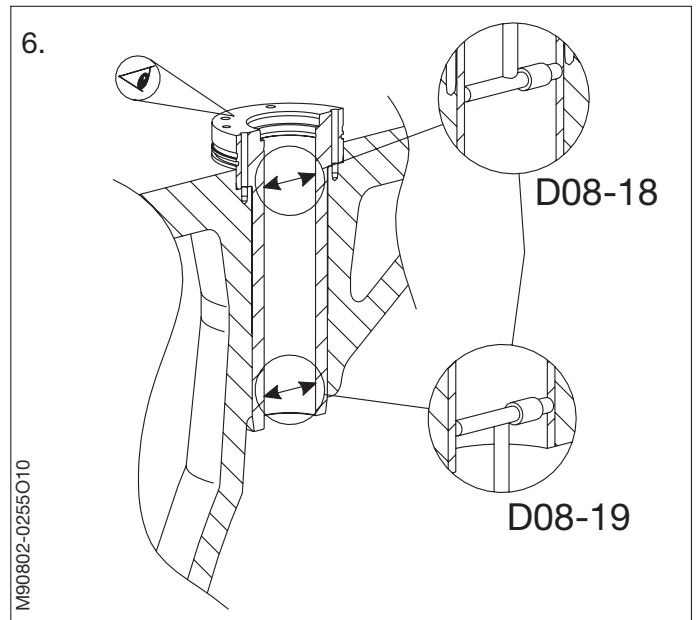
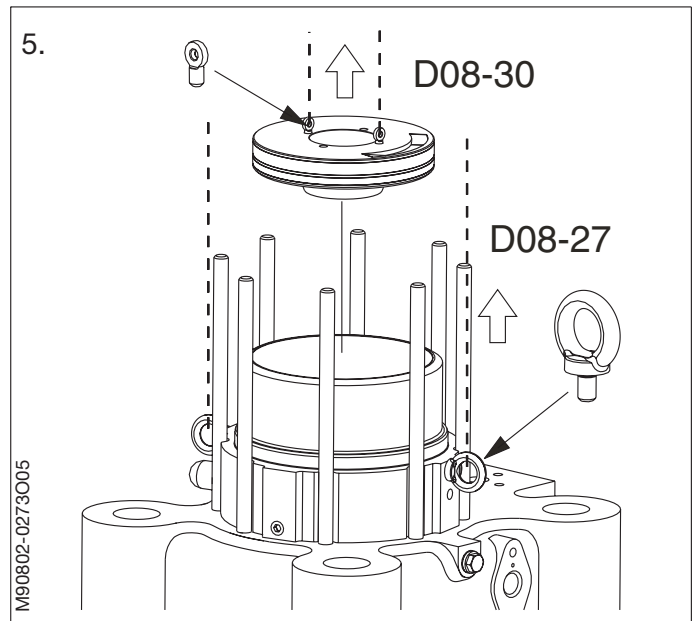
Mount two eye bolts in the air cylinder and dismount the air cylinder.

Remove and discard the O-ring on the air cylinder.

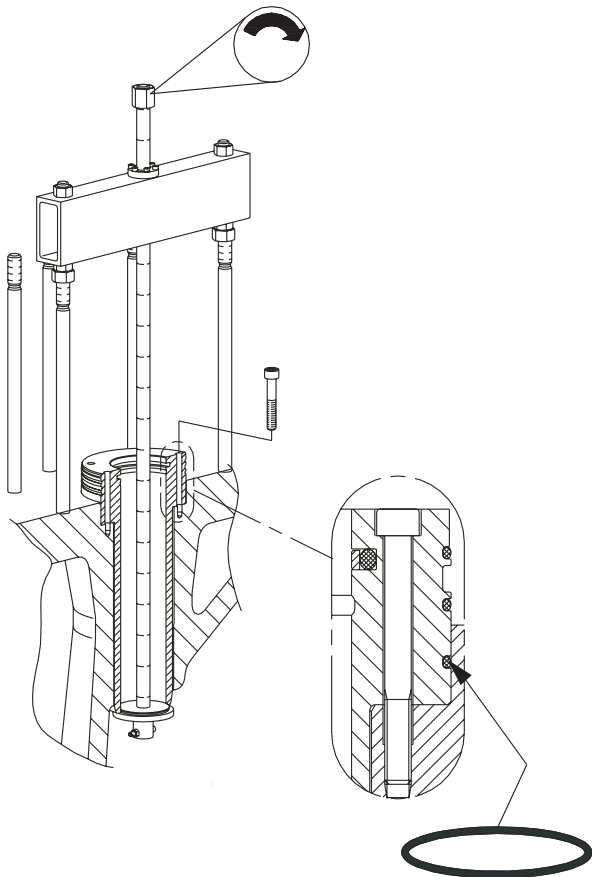
6. Guide bushing

Clean the exhaust valve housing.

Inspect the bushing in the spindle guide for wear and measure the top and bottom diameters. See *Data*.



7.

**7. Bushing renewal**

If it proves necessary to replace the bushing, remove the screws and knock out the bushing with a hammer and a suitable mandrel.

Alternatively, if the extractor tool is available, mount the extractor tool and pull up the bushing.

The extractor tool is optional, and can be ordered as spare part.

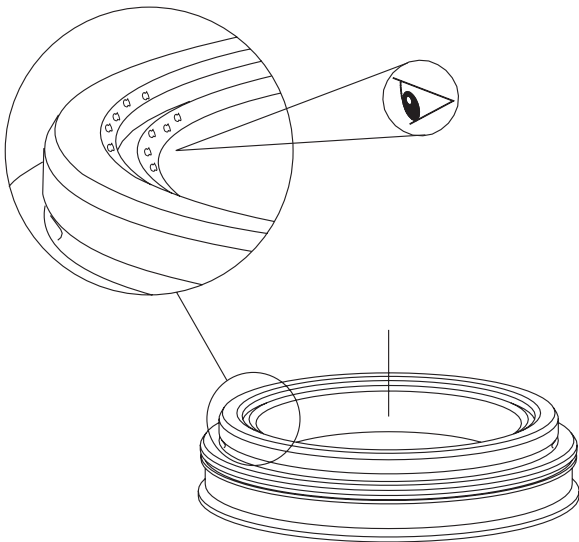
When a new bushing is mounted, mount a new O-ring for sealing between bushing and exhaust valve housing. Mount and tighten the screws for the bushing.

8. Bottom piece inspection

Before cleaning the bottom piece, See *Volume I, supplement 707-03*

Inspect and evaluate the seating according to Volume I.

8.



9. Bottom piece Measuring

In connection with evaluation of the bottom piece according to Volume I, wear or the total amount the seat has been ground, is measured as described below.

Use the bottom piece template and a feeler gauge, take measurements **G2** and **G3**. All measurements should be taken at four diametrically opposite points on the circumference of the bottom piece seating.

For maximum wear or grinding, see Data D08-11 and D08-12.

If the seat or the recess G3 is ground, see Data D08 – 09 for correct grinding angle.

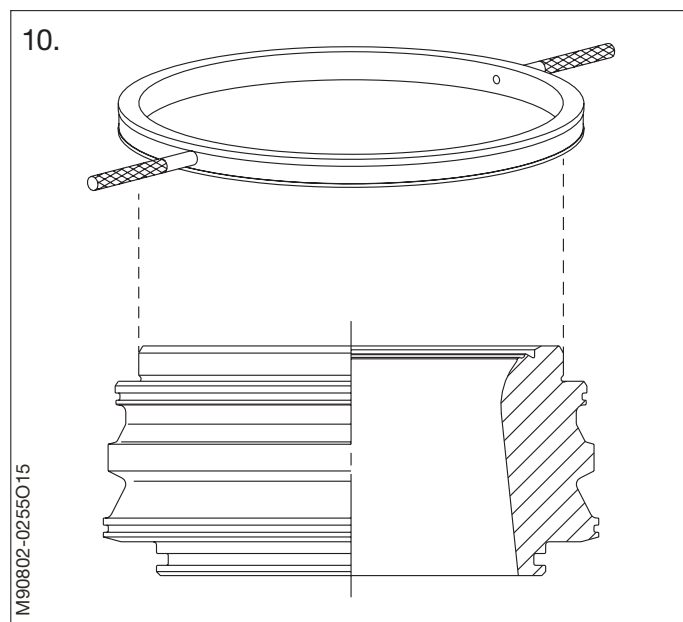
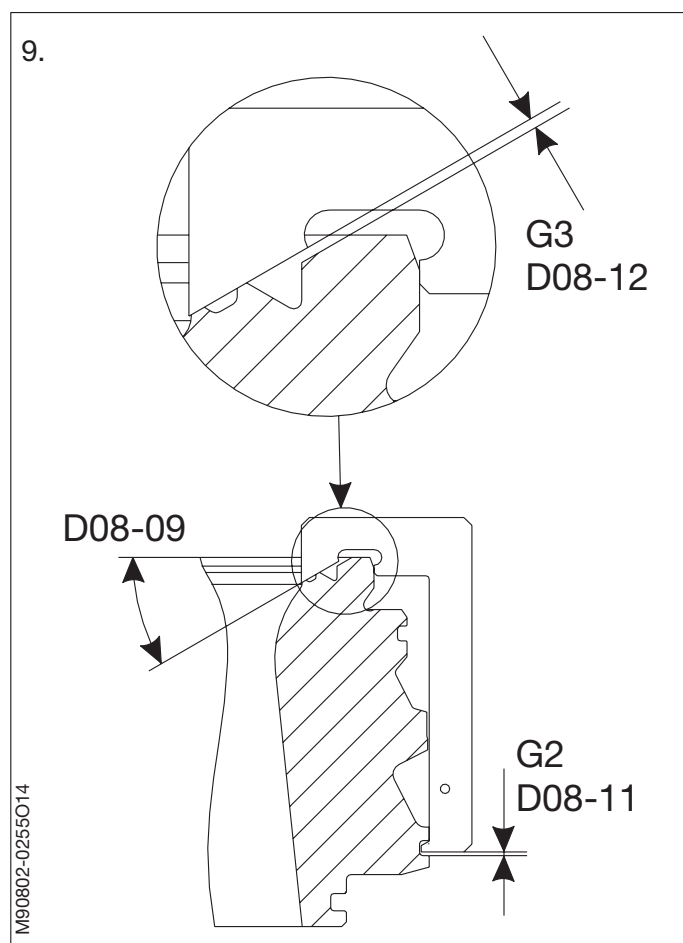
IF the template rests on the lower surface in the groove, i.e. **G2** = 0, THEN Further grinding or reconditioning is not recommended.

10. Bottom piece sealing face

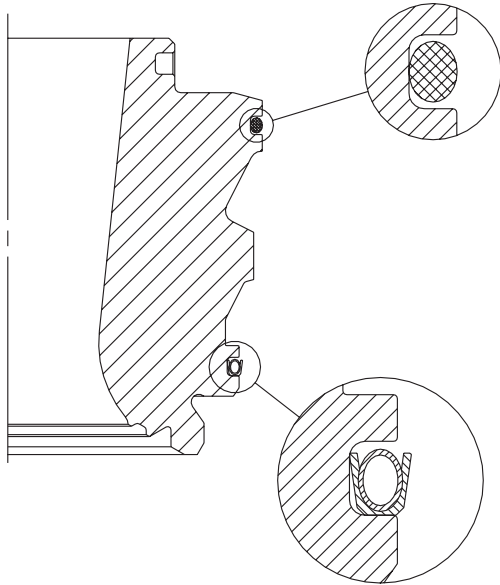
If necessary grind the outside seating.

Use carborundum 200 and the special grinding tool. Turn the special grinding tool **by hand**, turning alternately clockwise and anticlockwise. Grind until a smooth surface is achieved.

Clean the bottom piece.



11.

**11. Bottom piece sealing rings**

Mount new sealing rings on the bottom piece.

Mount one O-ring in the upper groove.

Make sure that the groove for the U-seal is clean.

Heat the U-seal in 100°C hot water for minimum five minutes and then mount it in the groove immediately.

12. Bottom piece mounting

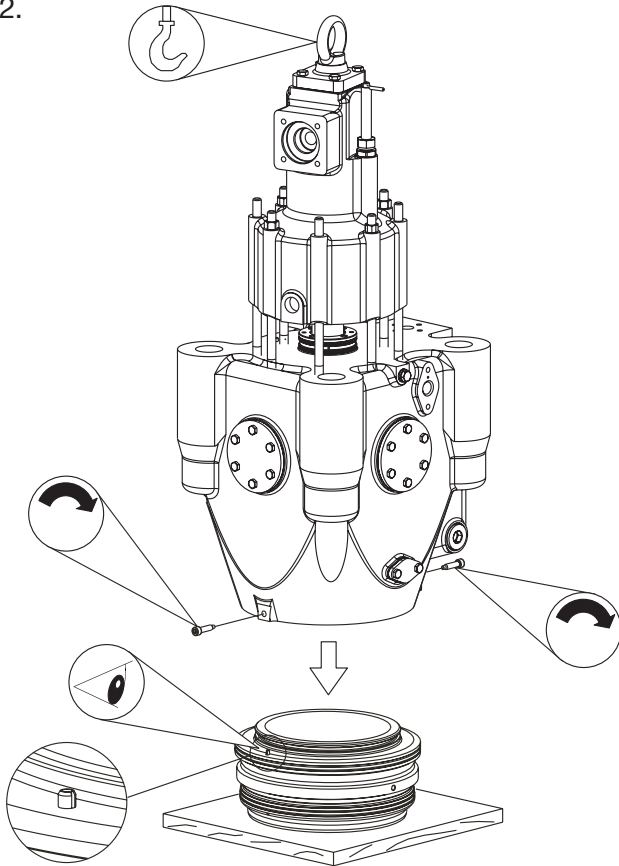
Use the oil cylinder as a lifting tool for the exhaust valve housing.

Check the face on the valve housing is clean.

If fitted, check that the guide pin in the bottom piece enters the hole in the bottom of the exhaust valve housing.

Land the exhaust valve housing on the bottom piece. Mount and tighten the retaining screws for the bottom piece.

12.



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13. Spindle inspection

Before cleaning the spindle, See Volume I, supplement 707-03

Inspect and evaluate the spindle according to Volume I.

14. Spindle burn-off and grinding check

In connection with evaluation of the spindle according to Volume I, burn-off and wear or the total amount the seat has been ground, are measured as described below.

The burn-off **F1** of the valve spindle is checked by measuring along the spindle template from point **A** to point **D** and in point **E**.

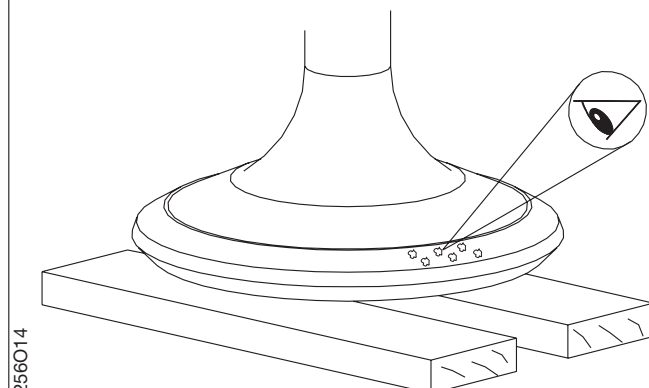
For the maximum allowable burn-off, see Data D08-13.

The maximum seat grinding, gap **G1** is measured between the spindle template, and the seating of the spindle with a feeler gauge.

For maximum seat grinding, see data D08 – 14.

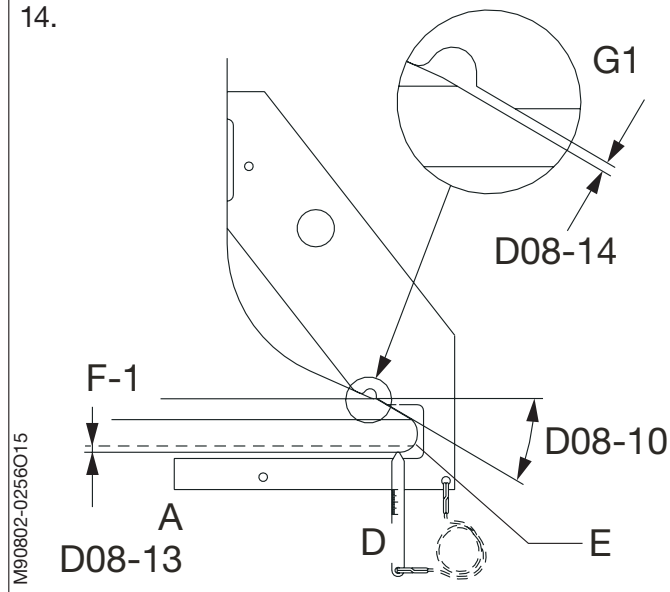
If necessary to grind the seat, see Data D08-10 for correct grinding angle.

13.



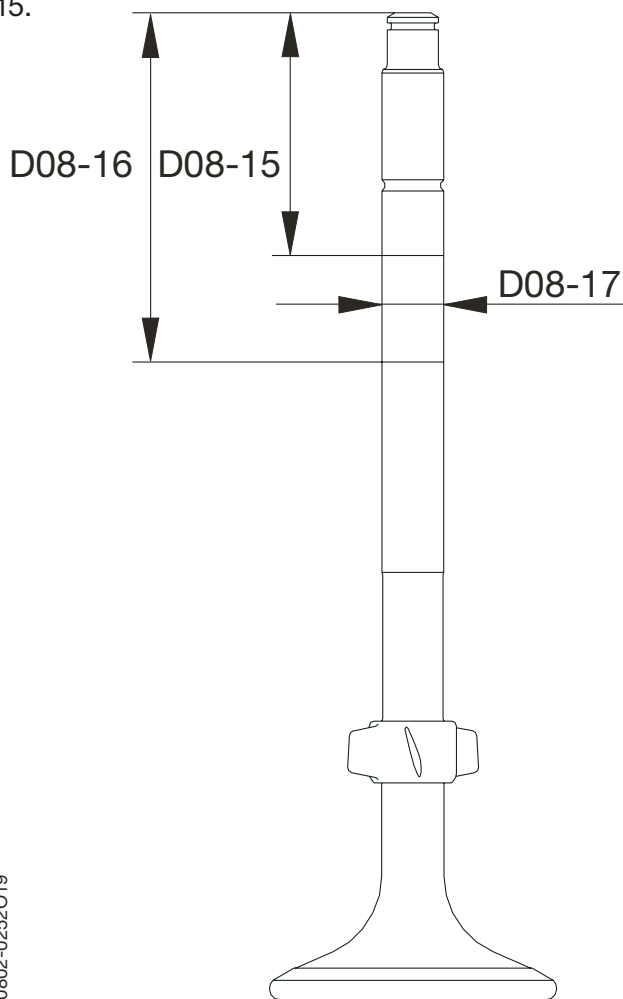
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14.



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15.



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15. Spindle stem check

In connection with evaluation of the spindle according to Volume I, the spindle stem wear is measured as described below.

Check the spindle stem for wear in the area D08-15 to D08-16 measured from the top of the spindle.

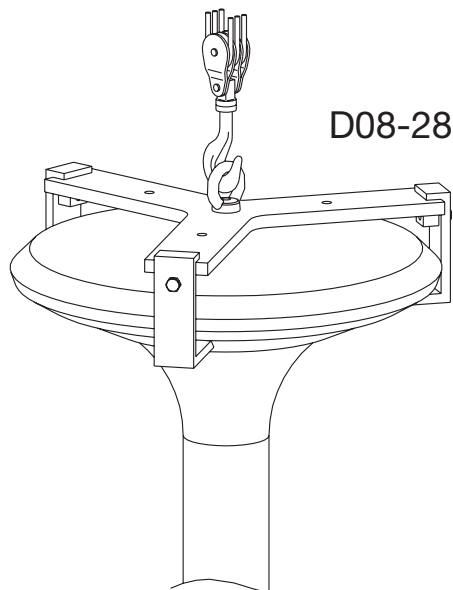
For minimum stem diameter, see data D08-17.

If the exhaust valve spindle needs reconditioning, contact MAN Diesel for advice.

16. Spindle transport

When lifting or transporting the exhaust valve spindle, use the special valve spindle lifting tool.

16.



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17. Spindle bushing sealing-rings

Check the top of the spindle bushing and the face on the housing is clean.

Fit new o-rings on the spindle bushing as shown in the sketch.

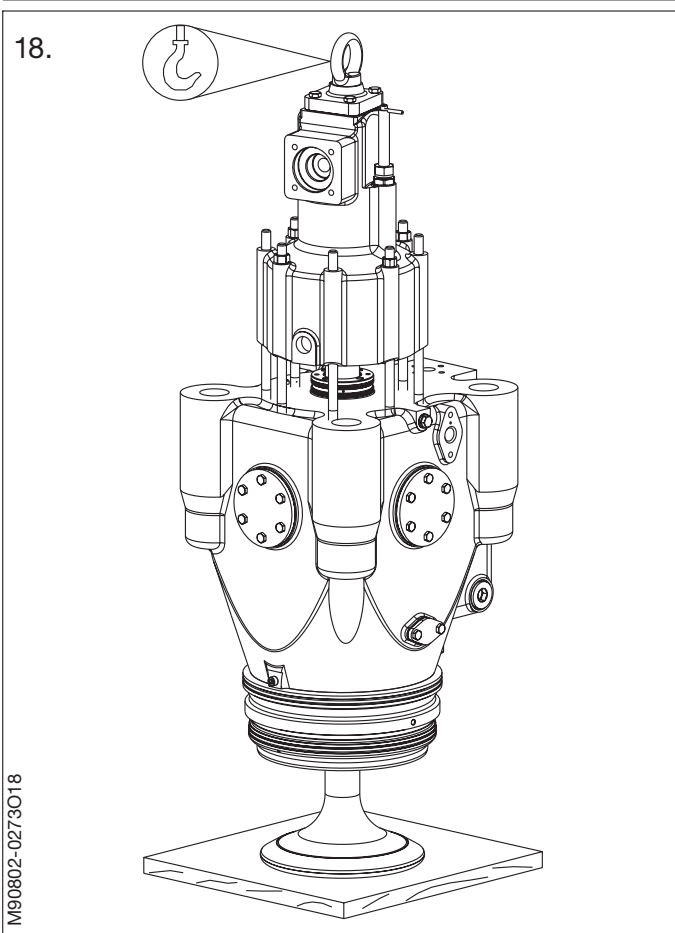
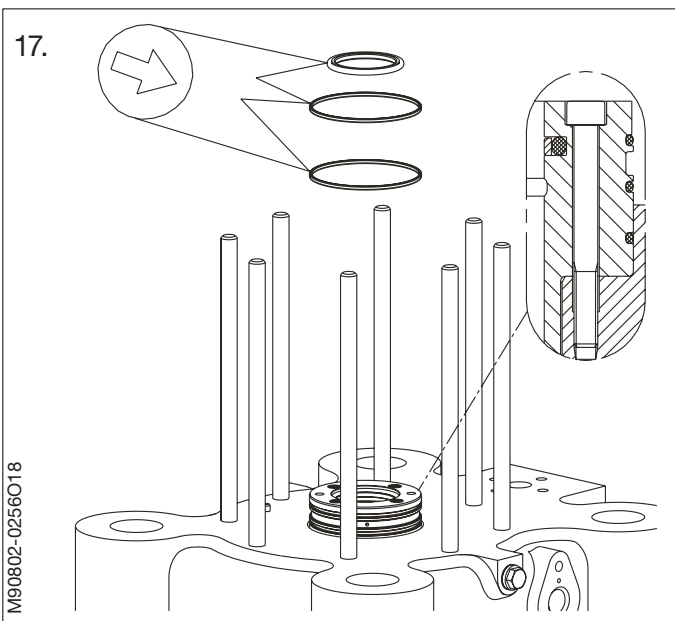
18. Spindle mounting

Lubricate the inside of the bushing in the spindle guide with plenty of lubricating oil.

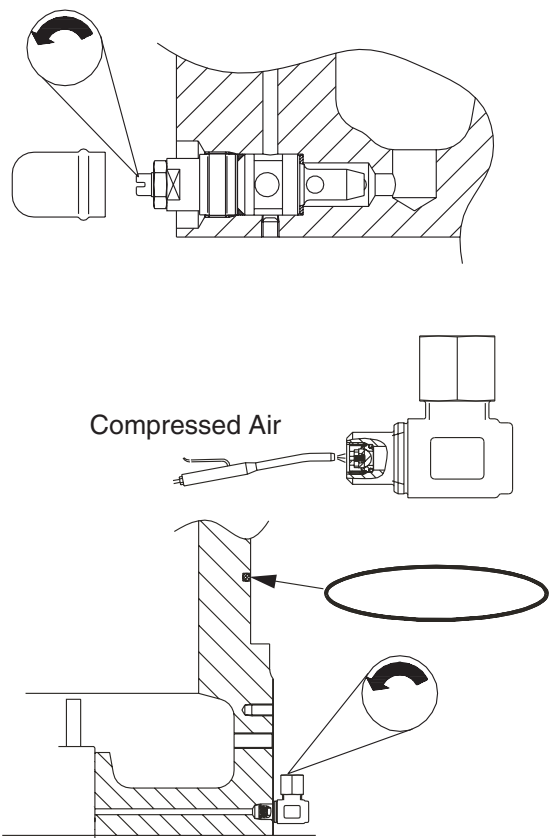
Use the oil cylinder as a lifting tool for the exhaust valve housing.

Carefully land the exhaust valve housing with the bottom piece on the exhaust valve spindle.

Remove the oil cylinder.



19.

**19. Air cylinder**

Clean the air cylinder.

Dismount the safety valve from the air cylinder, use the machined faces to unscrew the valve.

Unscrew the plug for the bore to the spindle stem bushing. Blow through all bores in the air cylinder. Clean the bottom of the air cylinder for possible coke deposits. Re-mount the plug. Mount a new O-ring in the groove on the outside of the air cylinder.

20. Safety valve

Place the safety valve in the pressure testing device. Connect the testing device with the high pressure pump by means of a hose.

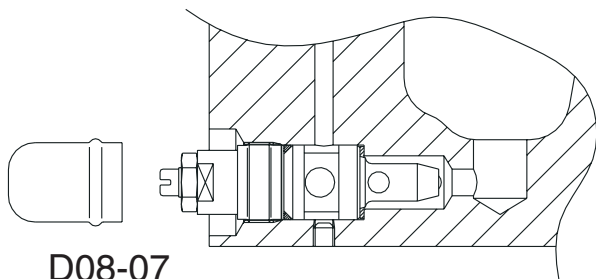
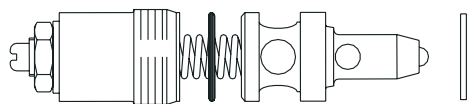
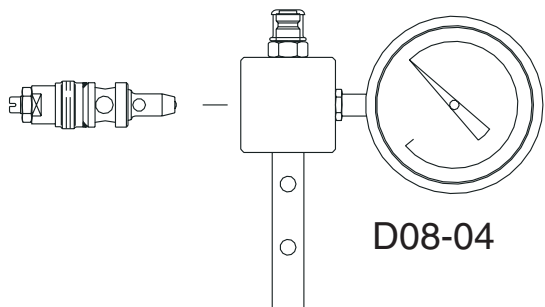
Check the opening pressure, see *Data*.

If the opening pressure is not correct, loosen the lock nut and, using a screwdriver, adjust the safety valve until the correct pressure is indicated. Tighten the lock nut and test the opening pressure once more.

Mount a new gasket and a new O-ring on the safety valve. Mount the safety valve in the bore near the bottom of the air cylinder. When mounting the safety valve in the air cylinder, use only the machined faces on the valve housing and tighten, see *Data*.

Finally, mount the protective cap.

20.



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M90802-0256021

21. Air cylinder mounting

Lubricate the O-rings on the spindle bushing.

Carefully land the air cylinder on the exhaust valve housing and check correct engaging with the locating pin. Some engines are not equipped with the locating pin.

Note!

Take care not to damage the O-rings on the outside of the spindle bushing.

22. Air piston seals

Check the teflon guide ring and teflon sealing ring for wear, if it is necessary to replace the sealing rings on the air piston, cut them and remove them.

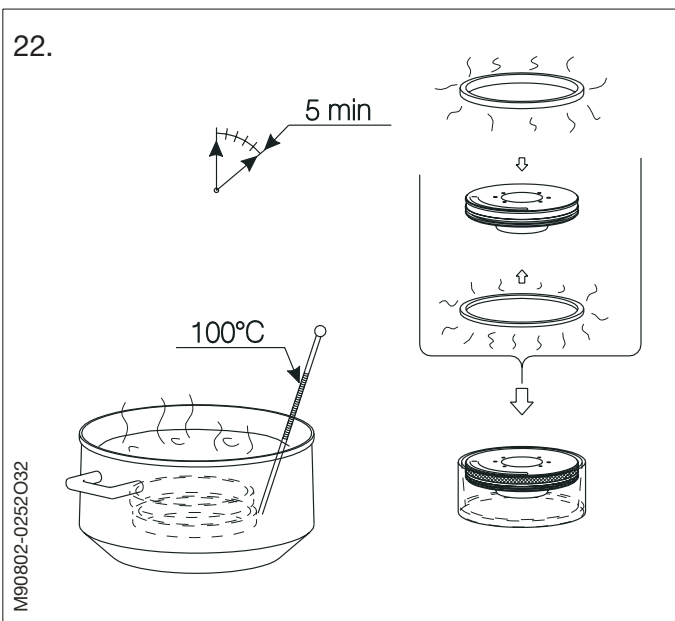
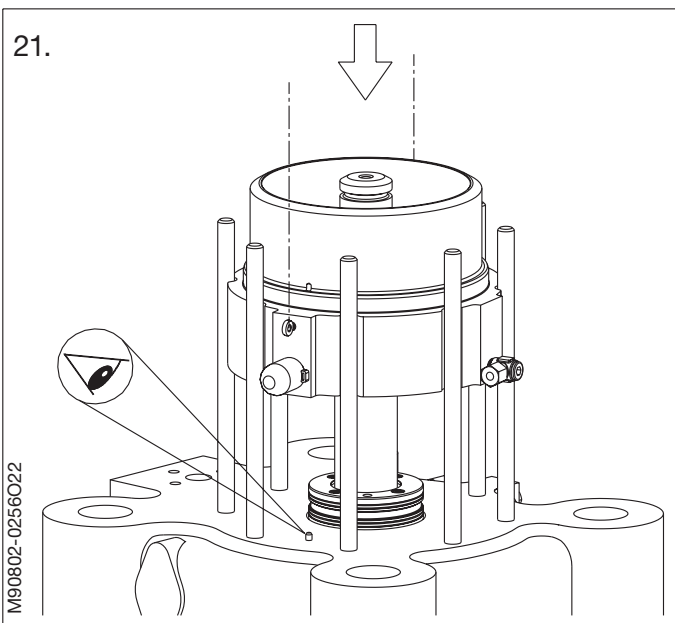
Be careful not to damage the edges of the ring grooves in the piston.

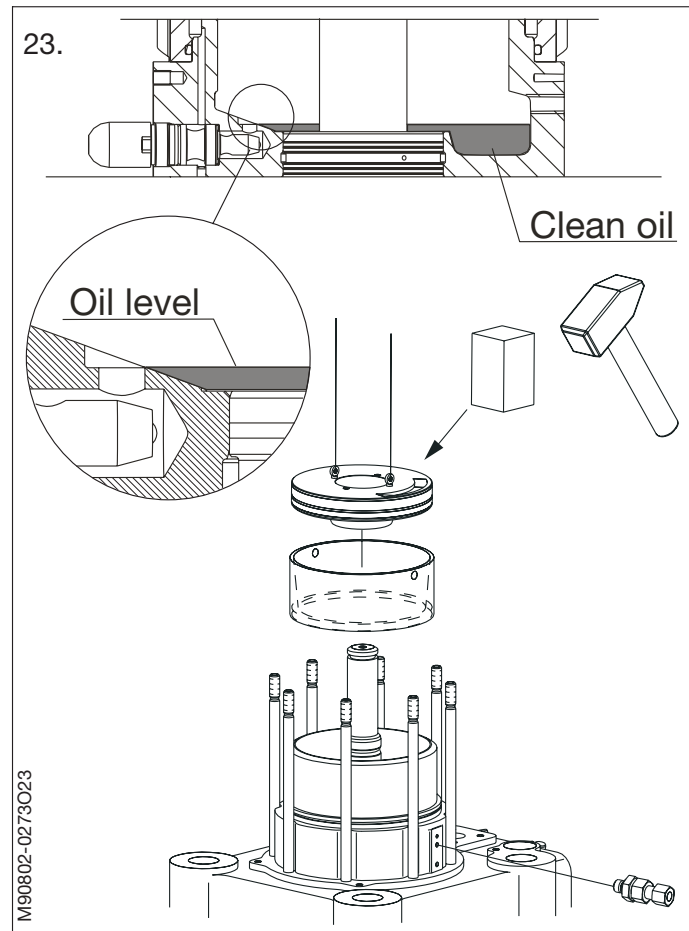
Note!

Before mounting, heat the new teflon rings in 100°C hot water for at least five minutes. When mounting the teflon rings, **be careful** not to damage the running surfaces.

After mounting, compress the teflon rings by means of the conical ring from tool panel 908. This is done by pressing the air piston through the conical ring.

Mount new O-rings in the inside grooves of the air piston.





23. Air piston mounting

IMPORTANT!

Fill the bottom of the air cylinder up to the drain hole with clean lubricating oil.

Fit the conical guide ring for the air piston on the air cylinder. Lubricate the guide ring and the running surface of the air cylinder. Lower the air piston over the valve spindle down into the air cylinder.

Use a nylon block and a hammer to press the air piston down in the cylinder

Remove the conical guide ring from the air cylinder.

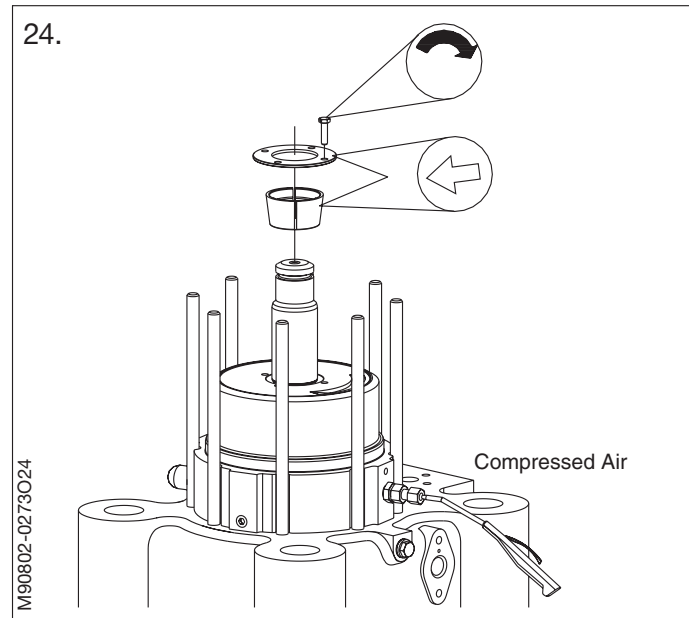
Mount the non return valve for spring air

24. Locking ring

Mount the two part conical locking ring and apply compressed air below the air piston to lift the piston.

Mount the flange on top of the air piston and, by means of the four screws, tighten the air piston and the flange together.

On valve models where the bolts are designed for securing, secure the bolts. See *Procedure 913-7*.



25. Oil cylinder

Loosen the screws and remove the lifting bracket from the oil cylinder to gain access to the orifice plug.

Remove and discard the gasket. Unscrew and clean the orifice plug. Remove and discard the sealing disc. Check and clean the bore for the orifice plug in the top of the oil cylinder.

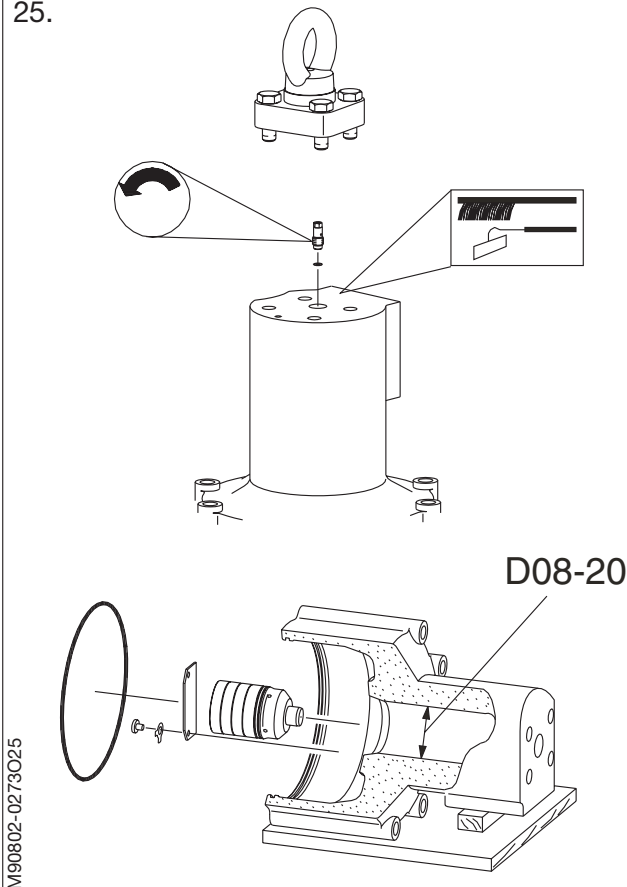
Place the oil cylinder in a horizontal position. Remove the flange and take out the piston. Inspect the inside of the liner in the oil cylinder for scoring. Measure the cylinder diameter. If the diameter of the oil cylinder exceeds Data D08-20, either renew the liner or send the complete oil cylinder to an authorised MAN Diesel workshop for overhaul, depending on the valve model.

26. Oil piston

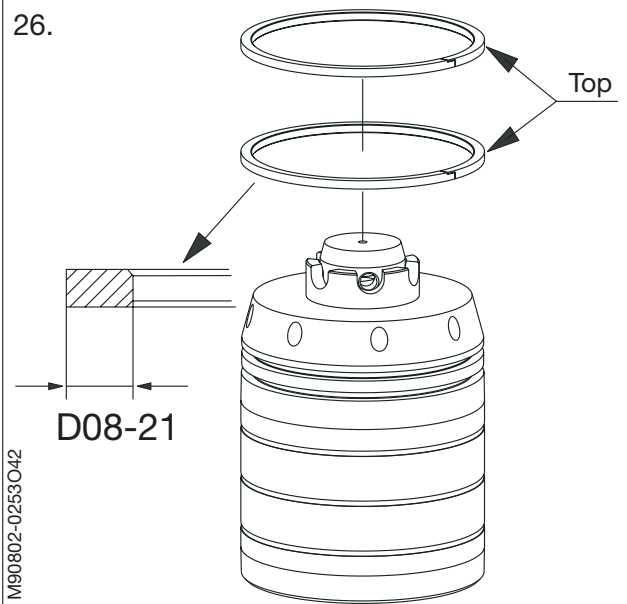
Remove the piston rings from the oil piston and check them for wear. If the thickness of the rings has worn down to the minimum, see Data, discard the rings and mount new ones. Check that the **TOP** mark on the piston rings faces upwards, when mounting

Check the mesh once more.

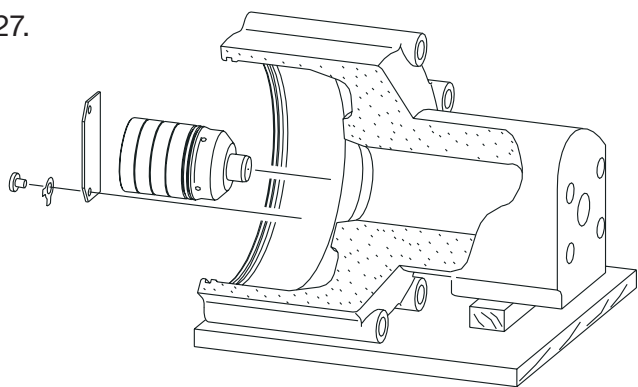
25.



26.

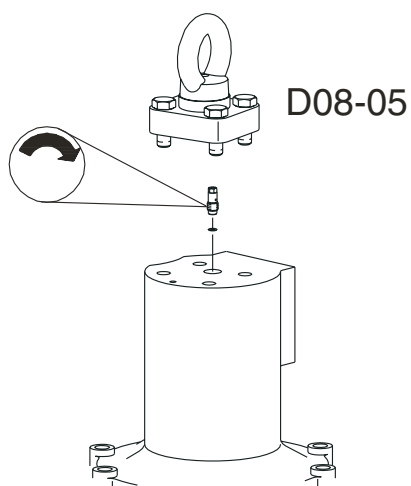


27.

**27. Oil cylinder assembly**

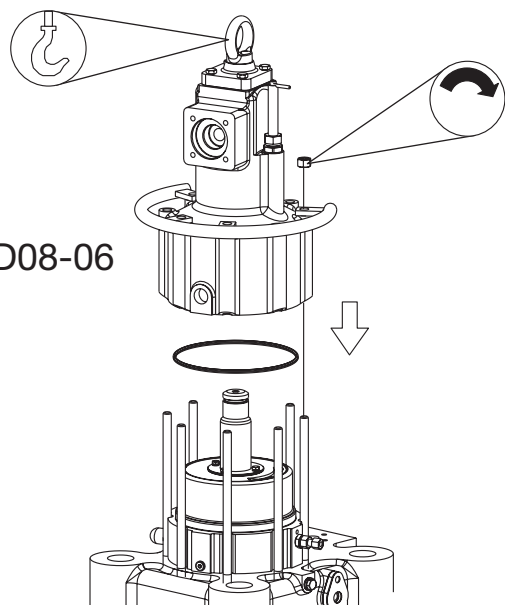
Mount the piston and the flange in the oil cylinder. Tighten and lock the screws.

Return the oil cylinder to an upright position. Mount a new sealing disc in the bore for the orifice plug. Screw the orifice plug into the bore. Place a new gasket on top of the oil cylinder. Mount the lifting bracket on the oil cylinder. Mount the screws and tighten to the torque stated in Data.

**28. Oil cylinder mounting**

Mount the overhauled oil cylinder on top of the air cylinder. Mount the safety strap and tighten the nuts. See Data.

28.



M90802-0273027

M90802-0273029

1. Checks before mounting

Before mounting an overhauled exhaust valve in the engine, it is recommended that the valve is checked and prepared as follows:

Check the oil level in the bottom of the air cylinder. Unscrew and remove the non-return valve and the safety valve.

Fill oil into the non-return valve hole until oil flows out from the safety valve hole. Refit the two valves again.

Connect compressed air to the air cylinder to close the valve.
Lift up the valve with the engine room crane.

Check that a 1.0 mm feeler gauge can be inserted about 15 mm into gap **G3**, to ensure that there is a clearance between the outer parts of the seating faces of valve housing and spindle.

The valve must be able to stay closed for a minimum of 15 minutes or according to the dropdown test in *Procedure 908-2.1*.

Note!

If the air cylinder is fitted with new sealing rings, the time the valve is closed may be considerably shorter than 15 minutes. The dropdown test should not be carried out until after a »running-in« period of 500 hours on valves with new sealing rings.

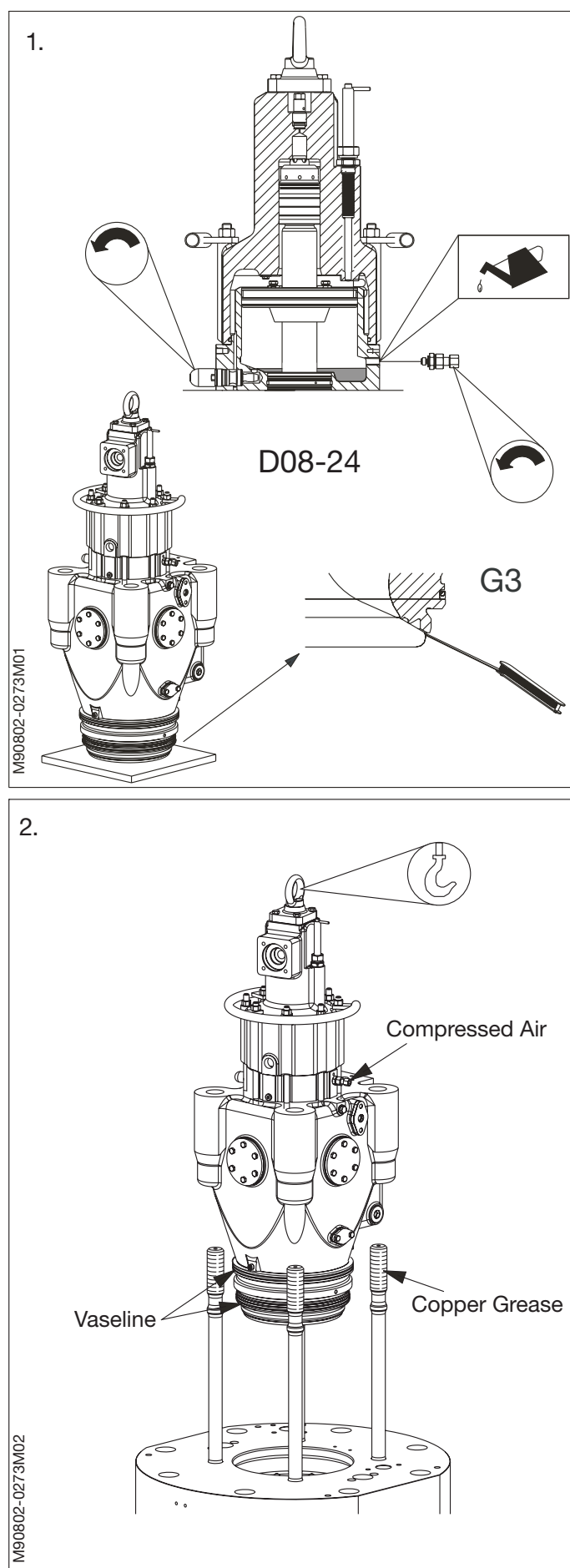
Check the valve is fitted with new sealing-rings in the grooves on the bottom piece.

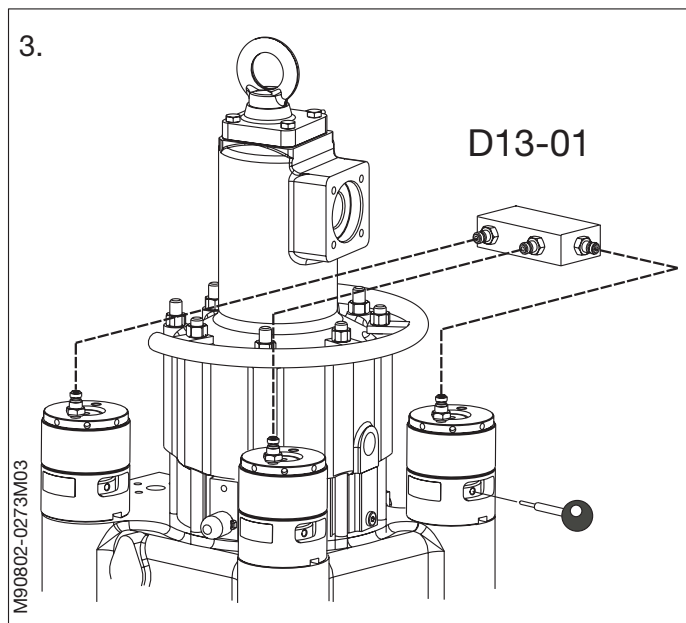
2. Exhaust valve landing

Close the valve with compressed air.

Lubricate the sealing rings with vaseline and the threads of the studs with anti-seizure paste.

Position the valve in the cylinder cover bore, guiding it in accordance with the exhaust flanges and the guide pin on the exhaust side of the exhaust valve.





3. Mounting of hydraulic nuts.

Mount and tighten the hydraulic nuts, see *Data*.

See *Procedure 913-1*.

Mount the protective caps on the exhaust valve studs.

4. Mount the pipe connections

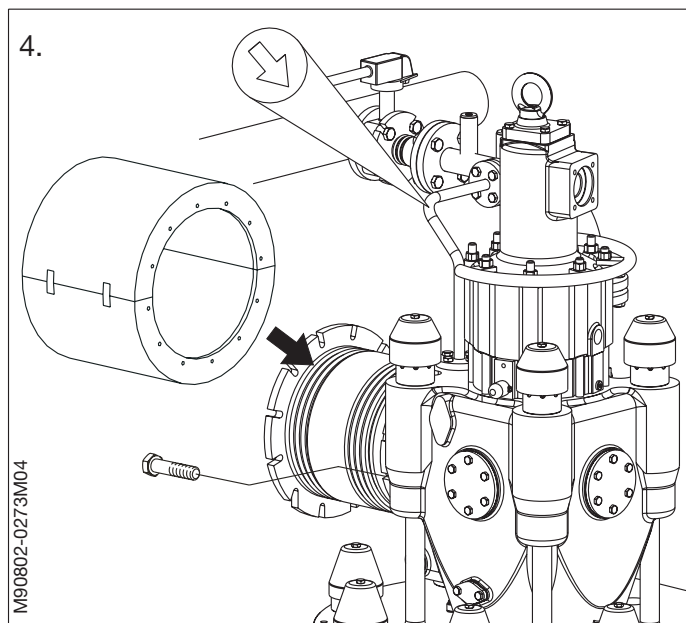
Connect the intermediate pipe to the exhaust valve.

Mount and tighten the screws to the exhaust valve housing.

Mount the insulation and plate jacket

for the intermediate pipe.

Mount the cooling water connections for the exhaust valve and the air pipe for the air spring.



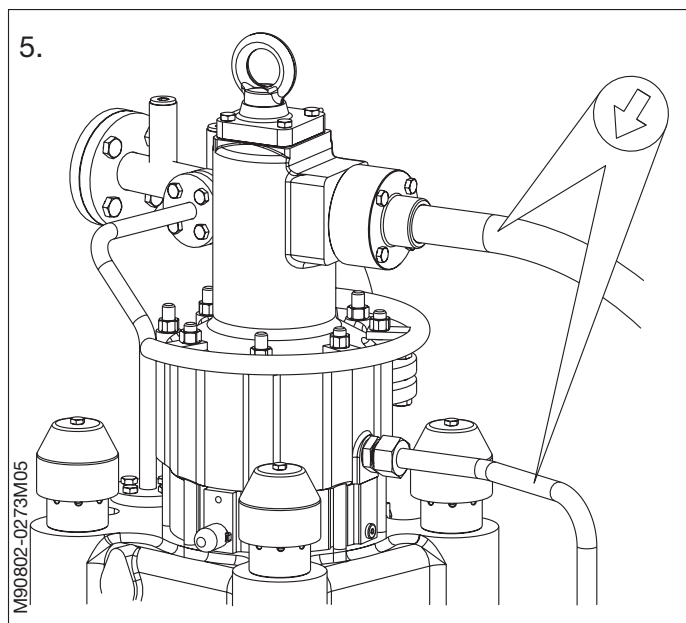
5. Mount the oil pipe connections

Mount the high-pressure pipe for the hydraulic valve actuation.

See *Procedure 908-1.2*

Mount the return oil pipe to the exhaust-valve.

Check that all pipe connections are properly tightened.



6. Open cooling water supply

Check that all water connections are mounted and tightened.

Open the cooling water inlet supply, and vent the exhaust valve, open the cooling water outlet valve

Open the air supply to the exhaust valve.

Check the tightness of the sealing ring between the bottom piece and the exhaust valve housing. See *Procedure 908-2.1*.

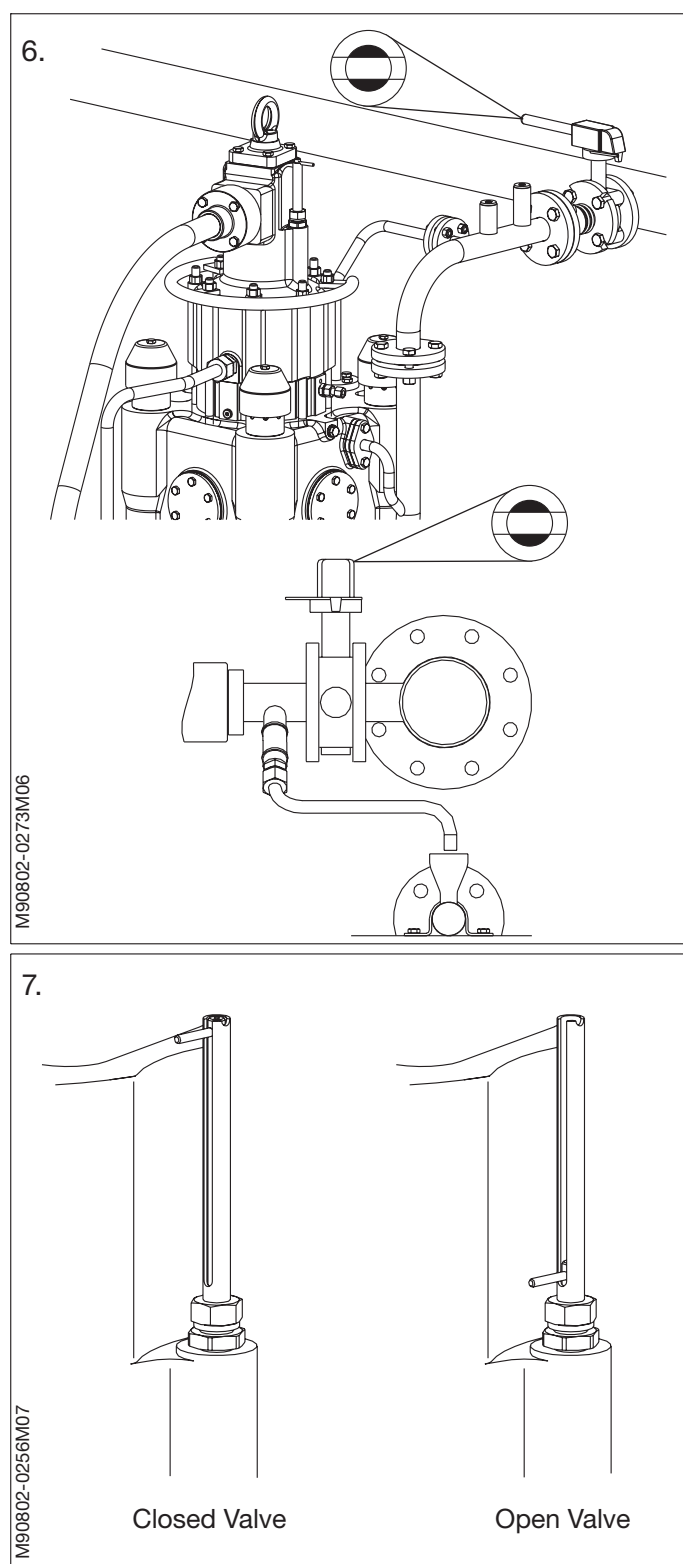
7. Air spring check

The air supply to the exhaust valve must always be connected and pressurized **before** turning on the oil supply to the exhaust valve actuator.

If the lubricating oil pumps have been started before the exhaust valve pneumatic springs are pressurize:

Engage the indicators on top of the exhaust valves and wait until the exhaust valves are closed.

Closing of the exhaust valves are eased if the lubrication oil pumps are stopped and the actuators are drained at the drain screws.



SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

☒	Stopped engine
☒	Shut off starting air supply – <i>At starting air receiver</i>
☒	Block the main starting valve
☒	Shut off starting air distributor/distributing system supply
☒	Shut off safety air supply – <i>Not ME engines</i>
☒	Shut off control air supply
☒	Shut off air supply to exhaust valve – <i>Only with stopped lubricating oil pumps</i>
☒	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☒	Stop lubricating oil supply
☐	Lock the turbocharger rotors

Data

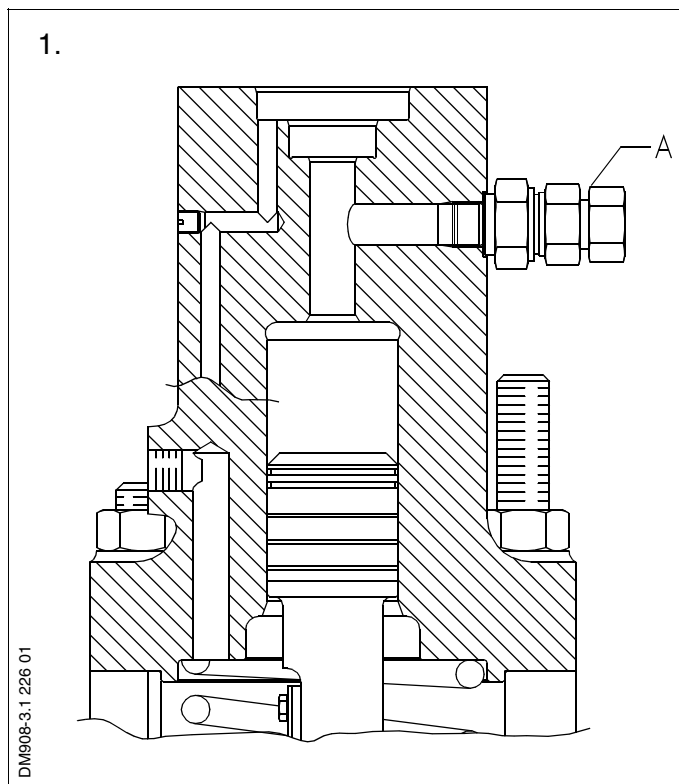
Ref.	Description	Value	Unit
D08-20	Oil cylinder max. inside diameter	71.2	mm
D08-21	Piston rings, min. thickness	2.6	mm
D08-32	Hydraulic actuator, tightening torque	300	Nm
D08-33	Locking plate, tightening torque	10	Nm
D08-34	Hydraulic actuator, tightening angle	25	°
D08-35	Hydraulic actuator	75	kg

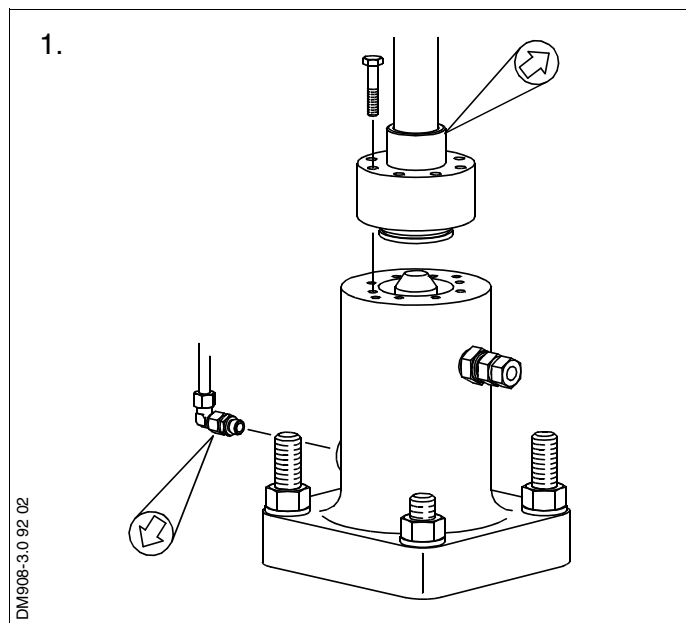
The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description

1. The non-return valve **A** is built into the oil inlet pipe to the actuator.

To check the valve, dismantle and clean it and connect a supply of working air (7 bar) to the outlet side of the valve (the side facing the actuator). If the air flow can be felt at the inlet side of the valve, the valve must be overhauled.





1. Stop the engine and shut off the oil supply.

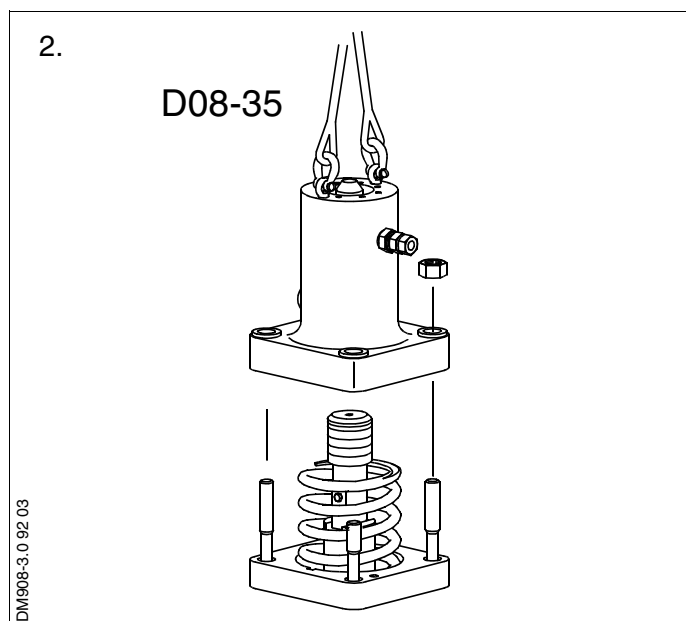
Turn the camshaft so that the roller rests on the circular part of the cam.

Remove the inspection cover of the roller guide housing to check the position of the exhaust cam.

Dismount the inlet pipe and the drain pipe from the exhaust valve.

Dismount the high-pressure pipe.
See Procedure 908-1.2.

2. Fit eye screws in the lifting holes of the actuator housing and hook on a tackle.



Unscrew the nuts on the two short studs, then loosen the nuts on the long studs successively until the roller guide spring is relieved.

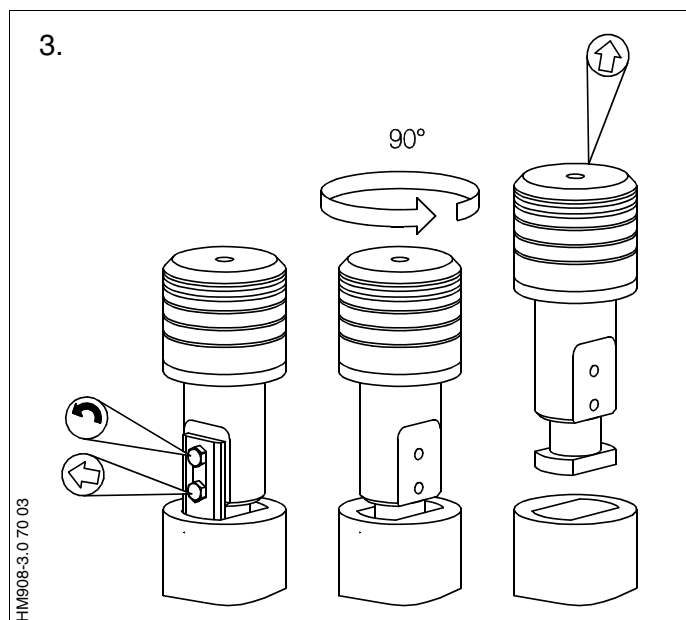
Dismount the nuts and lift the housing carefully, leaving the piston on top of the roller guide.

Remove the spring.

3. Remove the locking plate for the piston. Turn the piston **90°** to release it from the bayonet joint.

Remove the piston from the roller guide.

4. Overhaul the actuator housing and the piston as required.
See Procedure 908-3.3.



1. Clean the piston in kerosene and wipe dry with a clean piece of cloth.

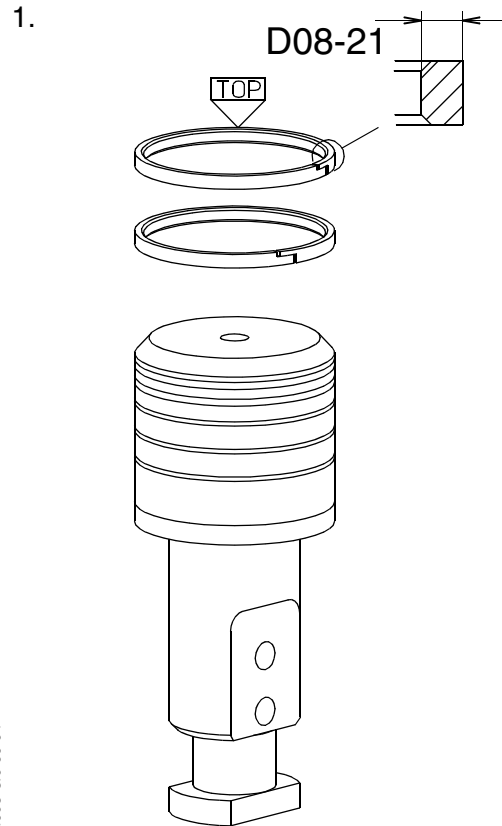
Take off the piston rings and check them for wear. If the ring thickness has worn down to the minimum (*see Data*), discard the rings and mount new ones.

Check that the **TOP** mark on the piston rings faces upwards when mounting.

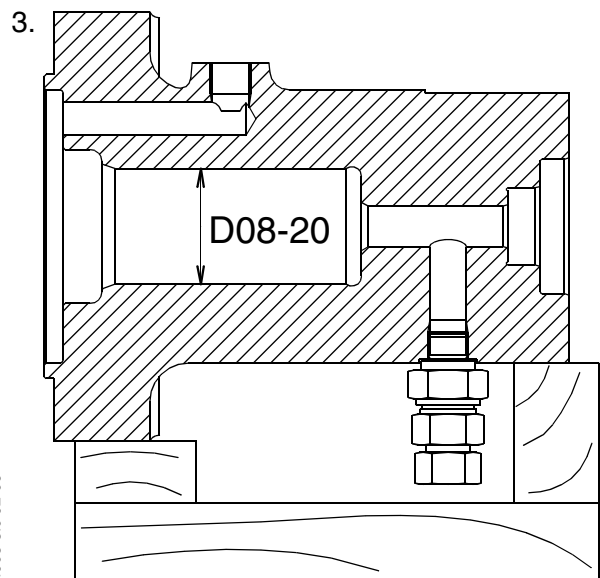
2. Inspect the sliding surfaces of the piston to ensure that there are no scratch marks or seizure marks.
3. Clean the oil cylinder and inspect the bores for deposits.

Check the bore of the oil cylinder for possible scores and measure it for wear.

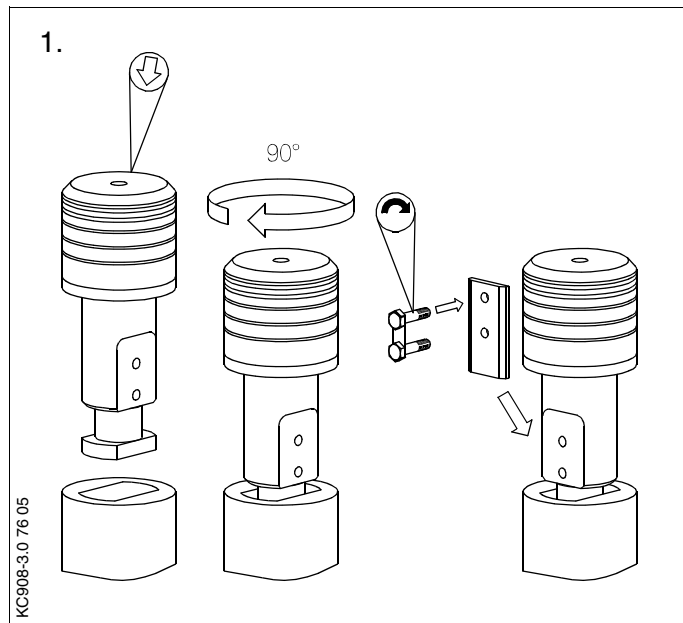
If the bore is seized or worn down to the measurement stated in Data, send the oil cylinder to an MAN B&W authorised workshop for reconditioning.



HM908-3.0 69 04



DM908-3.0 92 06



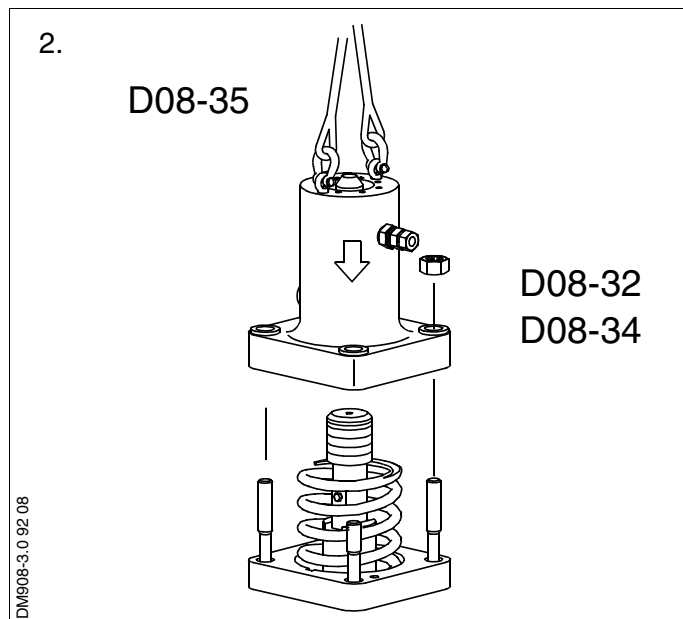
1. Mount the actuator piston in the roller guide and turn **90°** to lock it in the bayonet joint.

Mount the locking plate and tighten as stated in Data. Lubricate the piston with plenty of camshaft lubricating oil.

2. Mount the spring on the roller guide. Carefully lower the actuator housing on to the piston, taking care not to damage the piston rings.

Mount the washers and nuts on the two long studs and tighten alternately until the actuator housing rests firmly against the roller guide housing.

Mount the washers and nuts on the two short studs. Tighten all four nuts as stated in Data.

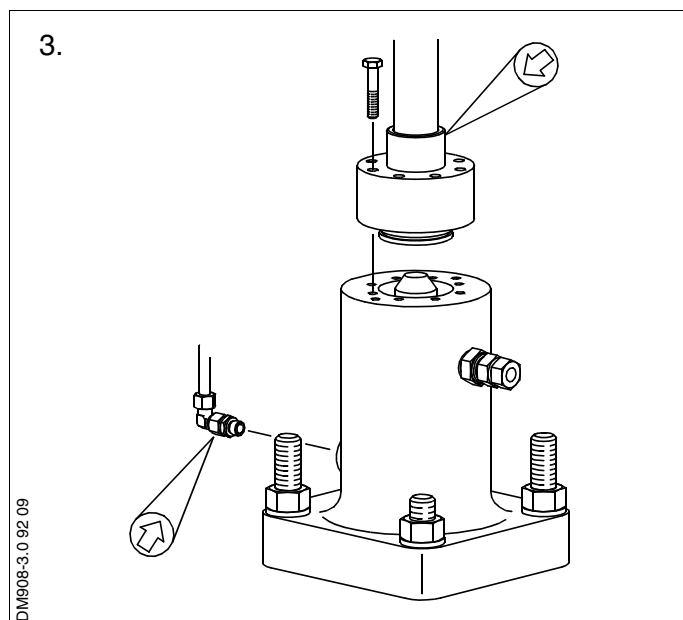


Note!

Use only the tightening angle **OR** the tightening torque, **NOT** both.

3. Mount the oil inlet pipe and the drain pipe from the exhaust valve.

Mount the hydraulic high pressure pipe. See *Procedure 908-1.4*.



SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

☒	Stopped engine
☒	Shut off starting air supply – <i>At starting air receiver</i>
☒	Block the main starting valve
☒	Shut off starting air distributor/distributing system supply
☒	Shut off safety air supply – <i>Not ME engines</i>
☒	Shut off control air supply
☒	Shut off air supply to exhaust valve – <i>Only with stopped lubricating oil pumps</i>
☒	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☒	Stop lubricating oil supply
☐	Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit
D08-36	Bushing/shaft pin, max. clearance	0.5	mm
D08-38	Roller guide complete	38	kg

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P90851	053	Lifting tool for roller guide
P90851	173	Lifting tool for exhaust valve roller guide
P91366	073	Dial gauge and stand tool

1. The exhaust valve roller guide can be partially checked while mounted on the engine, using the following procedure:

Lift the roller guide.
See Procedure 908-4.2.

2. Remove the camshaft inspection cover.

Turn the engine to provide the maximum space between the roller guide and the cam disc.

Check the movability of the roller by turning this by hand to ascertain if the roller moves freely and without unnecessary resistance or "hard" points.

Also inspect the surface of the roller for possible damage marks, seizures or scratches. Make sure that no traces of bearing metal exist in the roller guide housing.

Check the clearance in the slide bearing by lifting the roller relative to the roller guide and measuring the difference in distance between the cam disc and the roller in the upper and lower positions of the roller.

3. A more suitable and reliable method is to check the roller guide and the bearing clearance when the roller guide is dismantled.

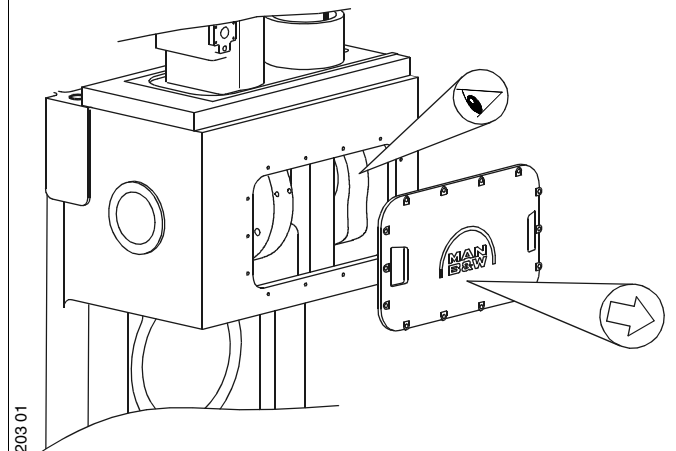
For dismantling of the roller guide see Procedure 908-4.2.

4. Inspect the surfaces of the roller guide and the roller for damage marks, seizures or scratches.

5. To measure the clearance in the roller guide slide bearing, place the roller guide vertically upside down on a couple of planks with the roller hanging freely.

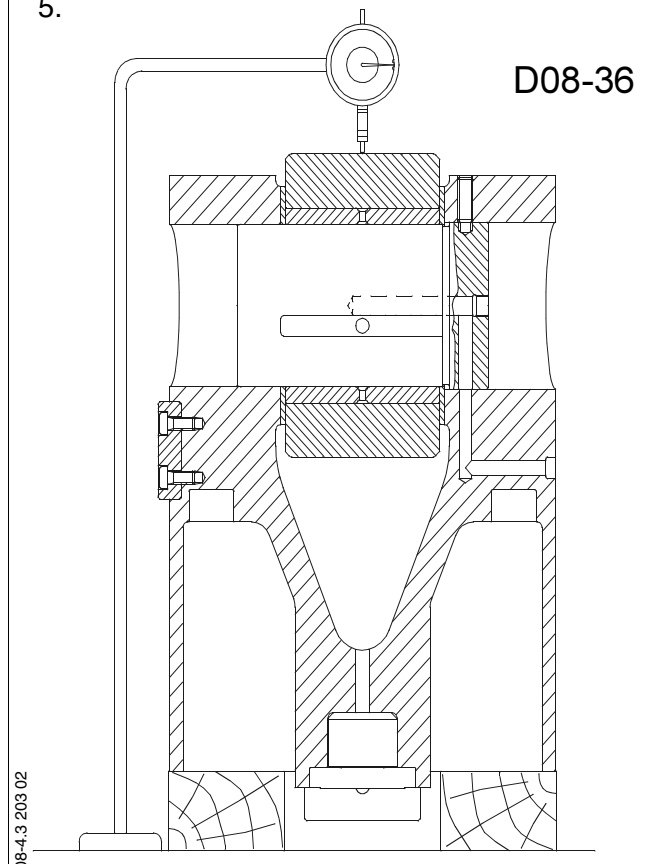
6. Place a dial gauge against the roller. Then lift the roller as much as the clearance permits, which makes it possible to read the clearance D-2 directly on the dial.

2.



GN908-4.1 203 01

5.



GN908-4.3 203 02

7. It is recommended that the roller guide should **ONLY** be dismantled if:

- irregularities when turning the roller,
- damage to the roller,
- larger clearance than stated on the Data sheet,
- seizure marks on the slide surfaces,

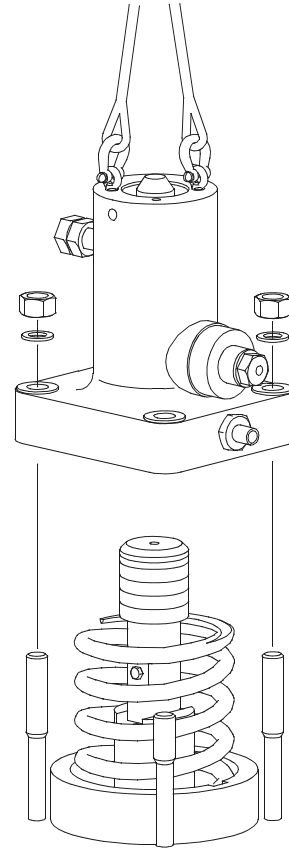
have been observed during the checking of the roller guide.

Note!
For disassembling of the roller guide, contact an MAN B&W authorized repair shop.

1. Dismantle the hydraulic actuator above the exhaust valve roller guide.
See Procedure 908-3.2.
2. Mount the roller guide lifting tool as follows:
 - Lift the lock ring of the tool up on the shaft of the tool.
 - Place the tool in such a manner that its foot rests on the thrust piece in the bayonet joint of the roller guide.
 - Turn the tool **90°** so as to allow the foot to engage properly in the bayonet joint.
 - Lower the lock ring, thereby securing the foot in the locked position.

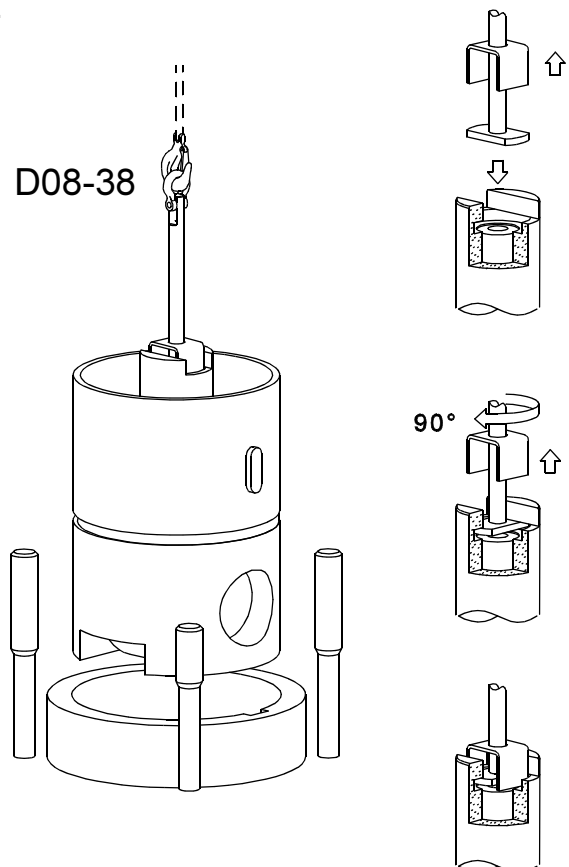
Lift the roller guide out of the roller guide housing, using the crane. Take care not to damage the sliding surfaces of the roller guide when lifting.

1.

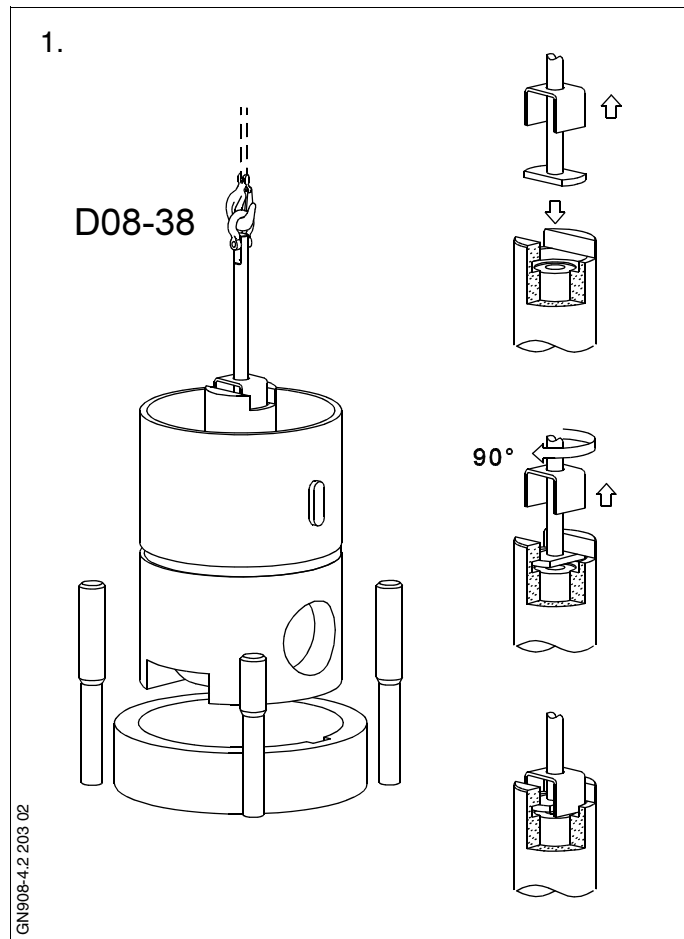


GN908-4.2 203 01

2.



GN908-4.2 203 02



1. Mount the lifting tool for the roller guide as follows:

- Lift the lock ring of the tool up on the shaft of the tool.
- Place the tool in such a manner that its foot rests on the thrust piece in the bayonet joint of the roller guide.
- Turn the tool **90°** so as to allow the foot to engage properly in the bayonet joint.
- Lower the lock ring, thereby securing the foot in the locked position.

Clean and lubricate the sliding surfaces of the roller and the roller guide and lower it carefully into the roller guide bushing.

Remove the lifting tool.

2. Mount the actuator.
See Procedure 908-3.4.

SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

☒	Stopped engine
☒	Shut off starting air supply – <i>At starting air receiver</i>
☒	Block the main starting valve
☒	Shut off starting air distributor/distributing system supply
☒	Shut off safety air supply – <i>Not ME engines</i>
☒	Shut off control air supply
☒	Shut off air supply to exhaust valve – <i>Only with stopped lubricating oil pumps</i>
☒	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☒	Stop lubricating oil supply
☐	Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit
D08-39	Check lifting height	10	mm

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P90851	173	Lifting tool for exhaust valve roller guide
P91351	010	Hydraulic pump, pneumatically operated
P91351	058	Hose with unions (3000 mm), complete
P91351	166	Angle union
P91351	225	Disc, round-plain, Cu ø11x3 mm
P91366	073	Dial gauge and stand tool

1. Turn the crank throw of the cylinder concerned in AHEAD direction until TDC. Check and write down the angle indicated on the turning wheel.

2. Remove the high pressure pipe, see procedure 908-1.2

Mount and tighten the lifting tool for the exhaust valve roller guide to the actuator piston (remove the screws for tightening to the actuator housing and do not lift the roller guide).

Mount a dial gauge (measuring range 0.00 - 20.00 mm) on the lifting tool for the exhaust valve roller guide.

Reset the scale and the pointer.

Press the dial gauge against the face of the actuator housing until the reading D08-39 + 1.00 mm is obtained.

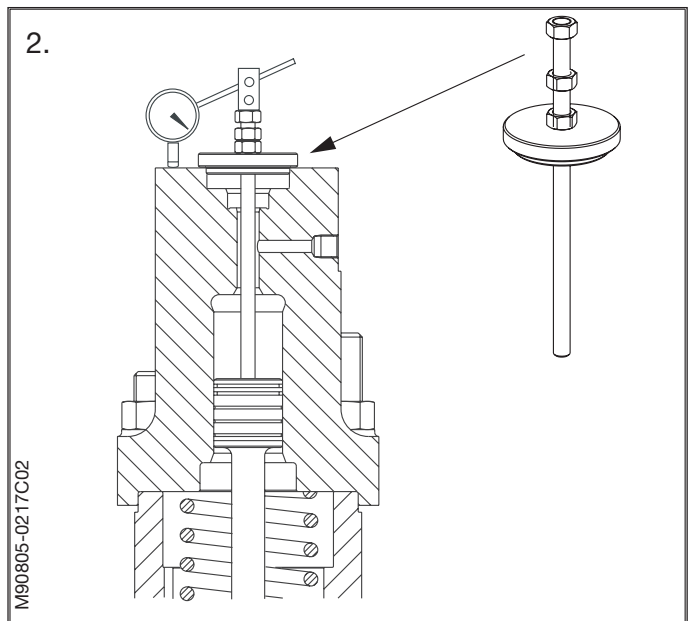
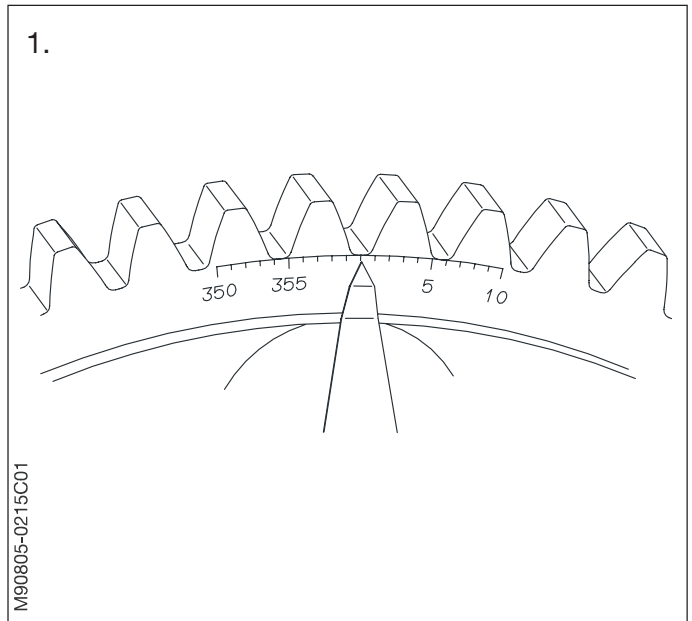
3. Turn the roller guide in AHEAD direction until lifted to D08-39, and the dial gauge gives the reading 1.00 mm and calculate the angle **A**, which is the number of degrees the crank throw has turned from TDC position to the present position.

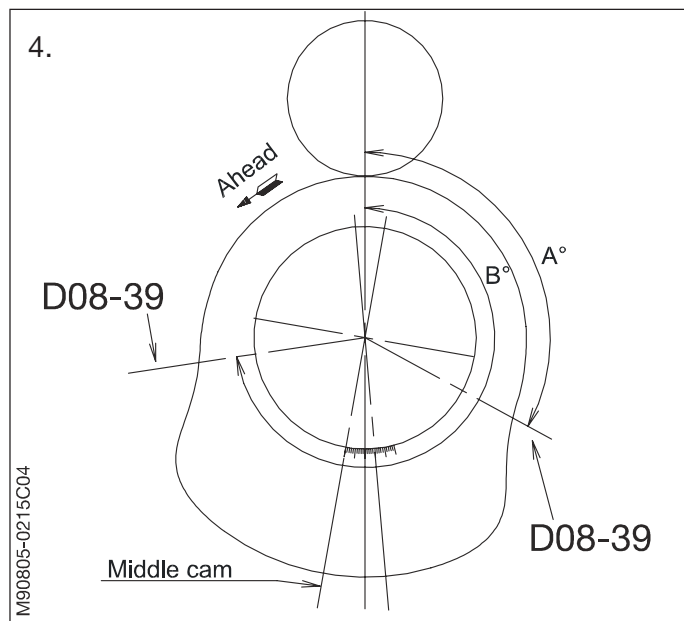
Continue turning AHEAD until the roller guide again is lifted to D08-39, and the dial gauge gives the reading 1.00 mm.

Calculate the angle **B**, which is the number of degrees the crank throw has turned from TDC position to the present position.

Note!

When calculating the angles **A** and **B**, remember to take into account that the scale on the turning wheel shifts from 360 to 0 (zero) degrees.





4. Calculate the lead angle as:
 $180^\circ - (A^\circ + B^\circ)/2$

The lead angle is normally a negative figure.

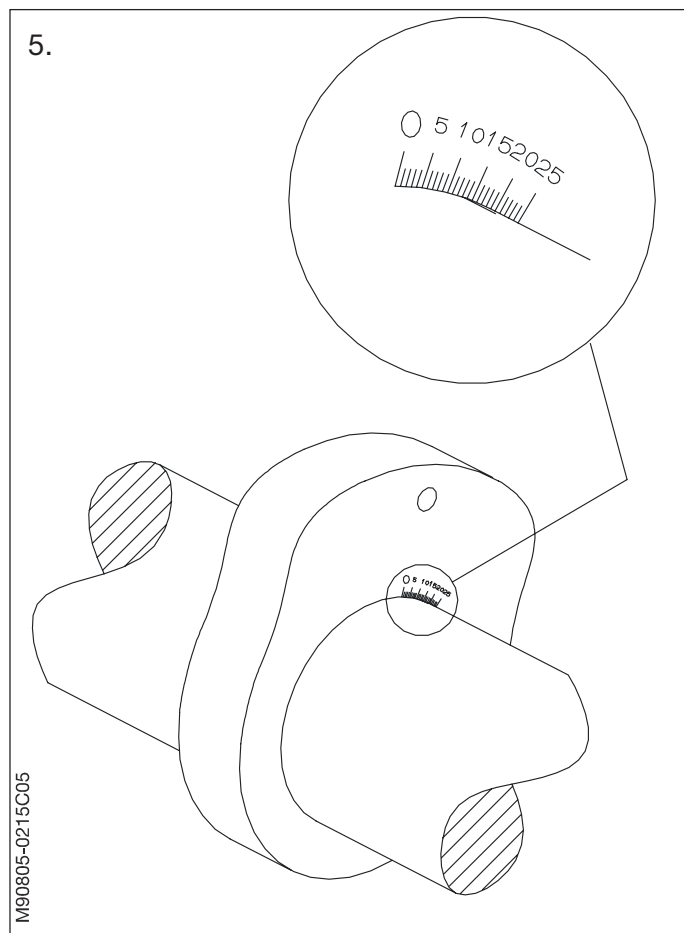
The illustration shows the position of the exhaust cam when the piston of the relevant cylinder is in TDC.

The position of the cam must be correct within a tolerance of $\pm 0.5^\circ$.

See the *Adjustment Sheet in Volume I, OPERATION, Chapter 701*, for the "lead angle" of the plant in question.

5. The cam lead angle can be checked visually by reading directly the scale position of the exhaust cam over the marking scratch on the camshaft and comparing this angle with the angle stated in the Adjustment Sheet.

At the same time, it is recommended to check the position of the camshaft with the pin gauge. See *Procedure 906-3.1*



Normally, adjustment of the exhaust cams should not be necessary. Before adjustment of any cams, see *Volume I, Chapter 706*.

1. Remove the inspection cover from the camshaft housing.
2. Dismount the exhaust valve high-pressure pipe, see *procedure 908-1.2*.
3. Turn the camshaft until there is access to the oil duct in the exhaust cam through the inspection hole.

If fitted, remove the plug from the oil duct (using, for instance, a screwdriver).

Insert three copper gaskets in the oil duct.

Mount one snap-on couplings in the oil duct, but do not tighten it.

Fit a hose between the snap-on coupling and the hydraulic high pressure pump.

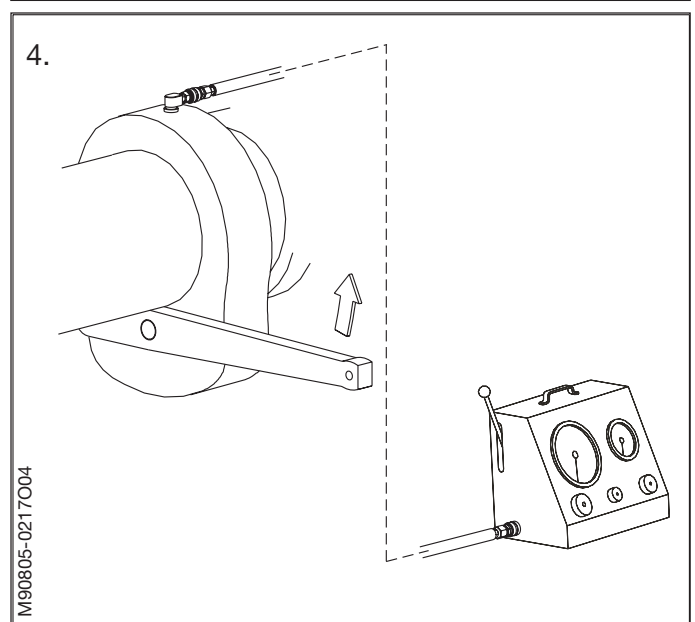
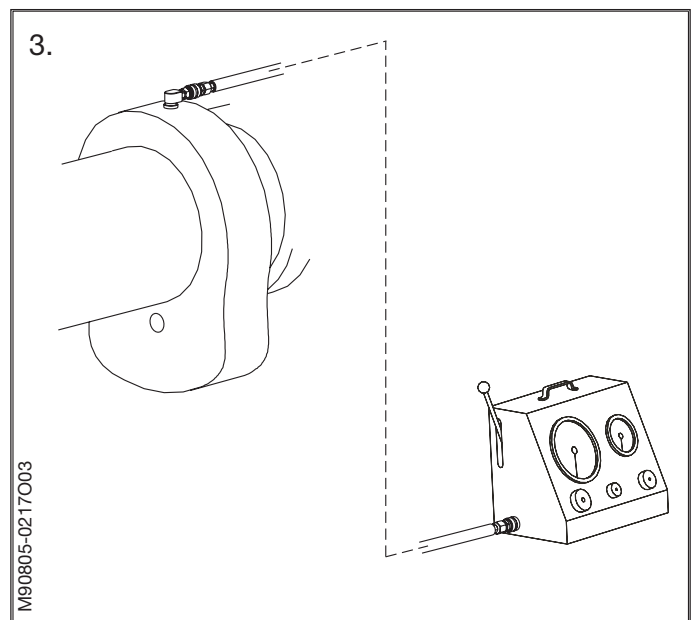
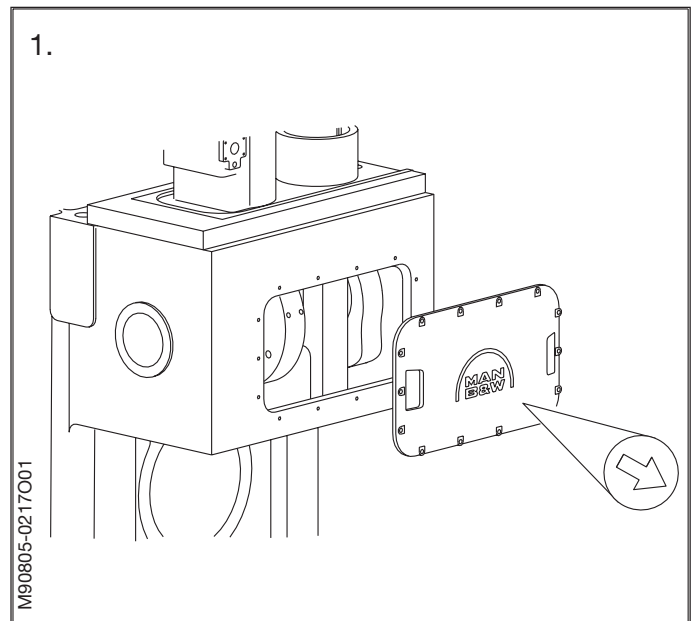
4. Mount the special spanner on the exhaust cam disc, ensuring that the two pins enter the holes in the cam.

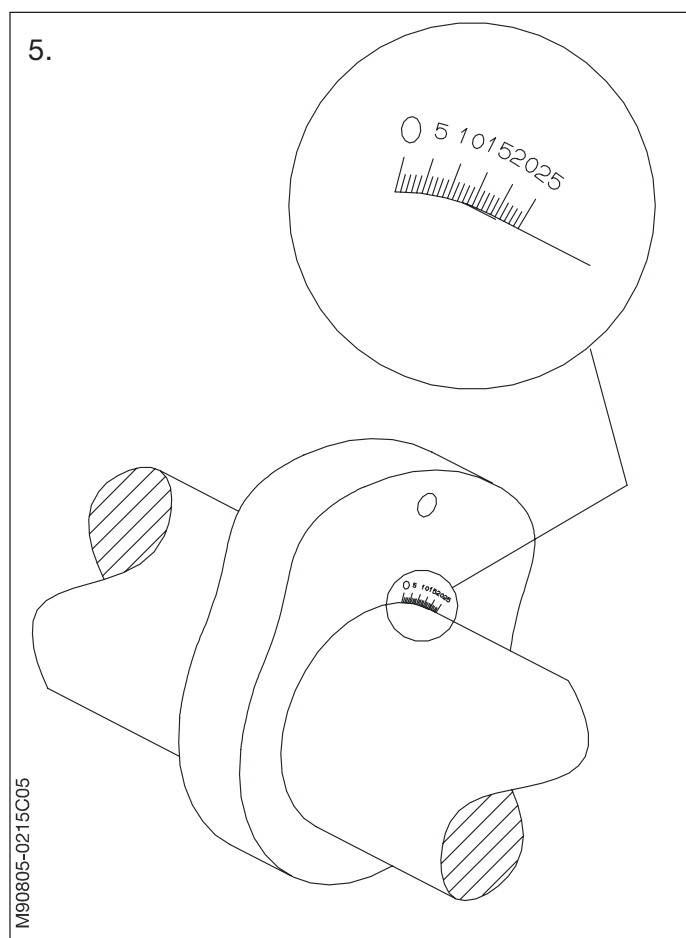
Apply a light pressure to the hydraulic system and, after venting the system, tighten the snap-on couplings.

Raise the hydraulic pressure until oil seeps out along the camshaft under the cam disc.

Keep the cam pressurised for at least two to five minutes before turning the cam. Oil must sweep out with the pump running during the operation.

Keep the cam pressurised during turning.





5. Turn the cam disc, using the fitted spanner, until the desired change of lead is obtained. Check the change on the cam scale

To increase the lead angle and reduce the compression pressure:

– turn the cam disc ASTERN.

To reduce lead angle and increase the compression pressure:

– turn the cam disc AHEAD.

6. After completing the desired turning of the cam disc, relieve the hydraulic system of pressure and dismount the spanner and the hydraulic equipment.

Wait at least 15 minutes – the cam must be allowed time to “settle” – before mounting the plug again in the oil duct of the cam disc.

After carrying out adjustment, measure the exhaust cam lead again. See *procedure 908-5.1*.

Write down and file the new results for the purpose of comparing with future measurements and adjustment.

7. Mount the inspection cover on the cam-shaft housing.
8. Mount the high pressure pipe for the exhaust valve. See *procedure 908-1.4*
9. After any adjustment of the cam position, p_{comp} must be recorded and compared with p_{comp} at a known engine reference load.

SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

☒	Stopped engine
☒	Shut off starting air supply – <i>At starting air receiver</i>
☒	Block the main starting valve
☒	Shut off starting air distributor/distributing system supply
☒	Shut off safety air supply – <i>Not ME-engines</i>
☒	Shut off control air supply
☒	Shut off air supply to exhaust valve – <i>Only with stopped lubricating oil pumps</i>
☒	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☒	Stop lubricating oil supply
☐	Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P90851	124	Tool for emergency open exhaust valve
P90851	173	Lifting tool for exhaust valve roller guide

Engine trouble might require that a cylinder be taken out of action. The engine can continue operation with the fuel and exhaust systems of a single cylinder disconnected. *See Volume I, Section 704.*

This can take place with the exhaust valve locked in either the open or the closed position. Either way, carry out the following procedure:

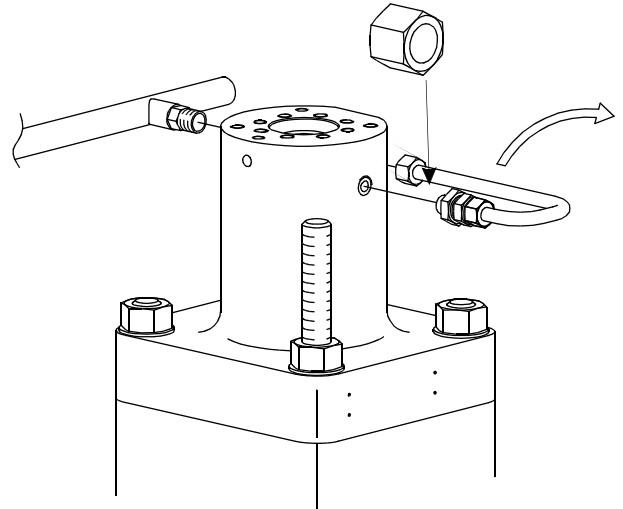
1. Lift the relevant fuel pump roller guide.
See Procedure 909-16.
2. Dismantle the hydraulic high-pressure pipe. *See Procedure 908-1.3.*
3. Dismantle the non-return valve on the oil inlet pipe. Loosen the union nut in the exhaust side of the oil pipe and turn the pipe to one side. Mount the non-return valve on the oil pipe and blank off the non-return valve with the plug from tool panel 908.
4. Remove the inspection cover of the camshaft housing. Turn the engine until the exhaust roller guide is at TDC.
5. Position the lifting tool on the actuator and screw the spindle into the top of the actuator piston.

Tighten the lowermost nut on the tool to pull up the roller guide as far as possible.

Tighten the upper nut of the tool against the lower nut to secure the roller guide in this position.

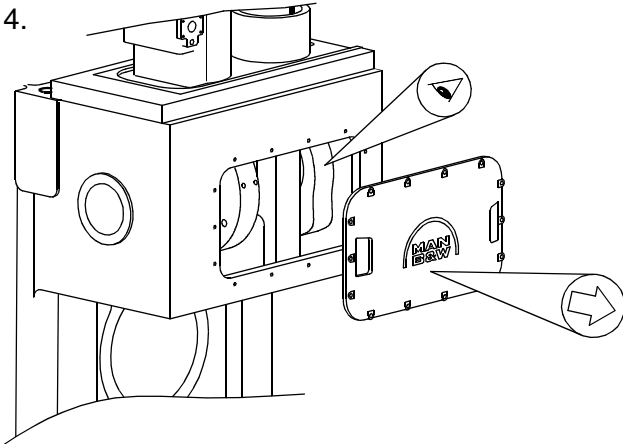
6. If the cylinder is to operate with the exhaust valve closed, the air supply must remain connected to the air cylinder of the exhaust valve. No further action is required.

3.



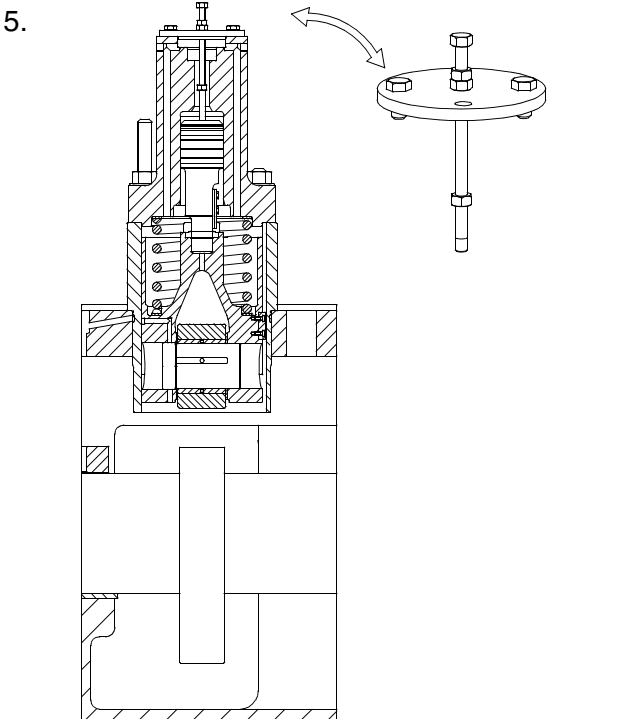
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4.



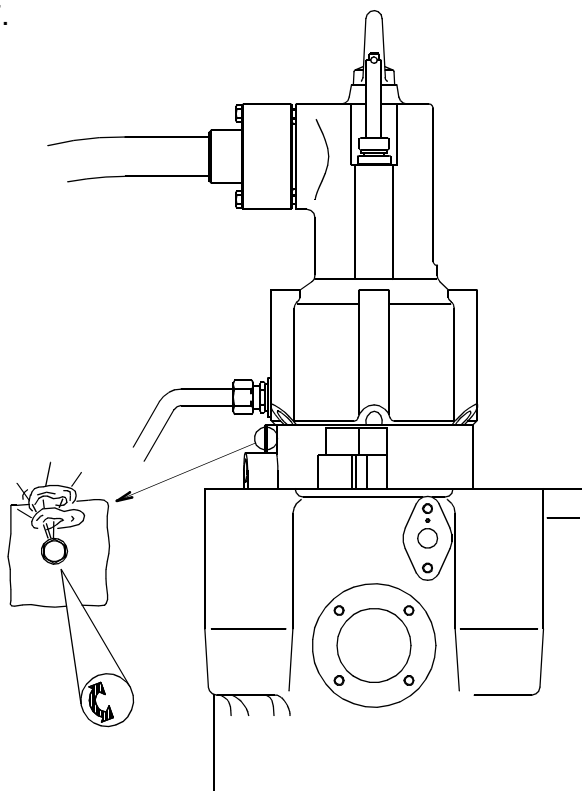
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5.



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7.



If the cylinder is to operate with the exhaust valve open, continue the procedure as follows:

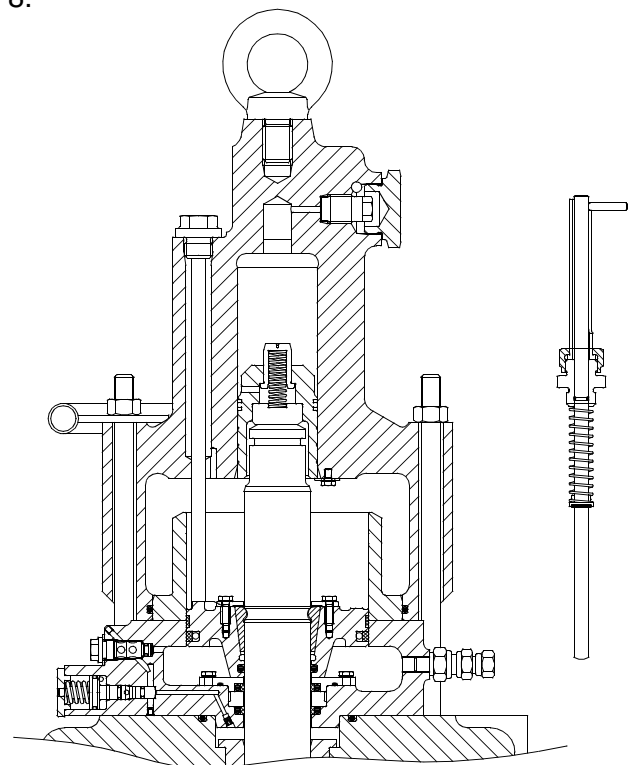
7. Turn off the air to the exhaust valve.

Remove the venting plug screw, fitted just above the ball cock on the air cylinder, to let the air inside the air cylinder escape.

8. Dismantle the rotation indicator on the oil cylinder and mount the tool for opening of the exhaust valve.

The tool will keep the piston of the air cylinder in the bottom position and thus keep the exhaust valve open.

8.



After overhaul of the cylinder unit, re-engage the exhaust valve gear as follows:

9. Loosen the nuts of the lifting tool to lower the roller guide on to the exhaust cam. Remove the tool.
10. If the engine has been running with the exhaust valve open, remove the tool mounted in the oil cylinder and mount the rotation indicator.

Note!

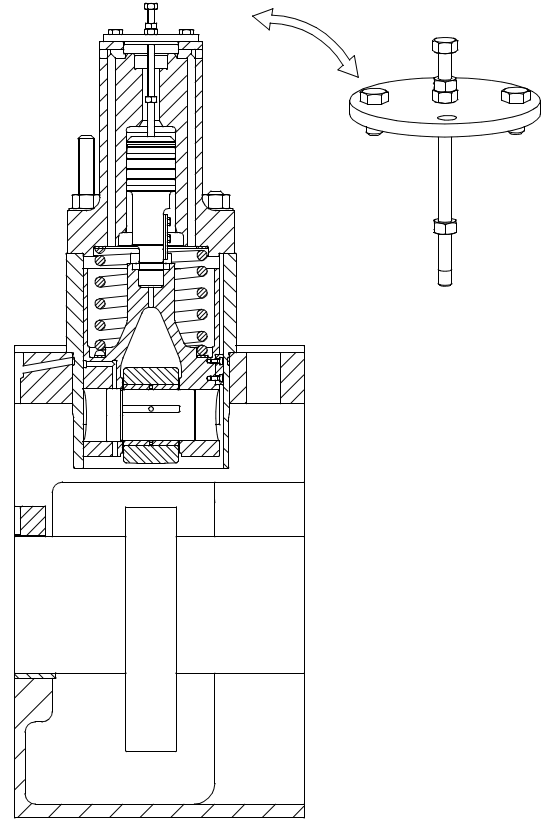
The rotation indicator should only be left in the engaged position for short periods of time during operation of the engine.

Leaving the rotation indicator in the engaged position continuously might damage the exhaust valve due to the formation of a wear groove in the air piston of the exhaust valve.

Re-connect the air supply to the air cylinder.

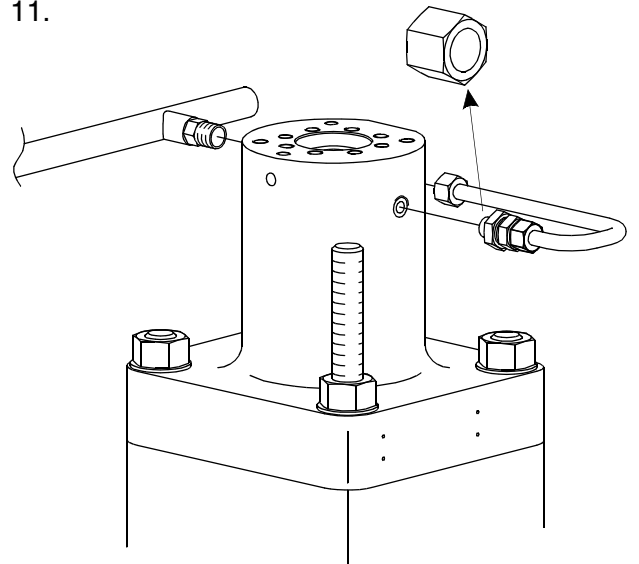
11. Remove the plug from the non-return valve and re-connect the oil inlet pipe on the actuator housing.
12. Mount the hydraulic high-pressure pipe.
See Procedure 908-1.2.
13. Engage the fuel pump roller guide.
See Procedure 909-16.

9.

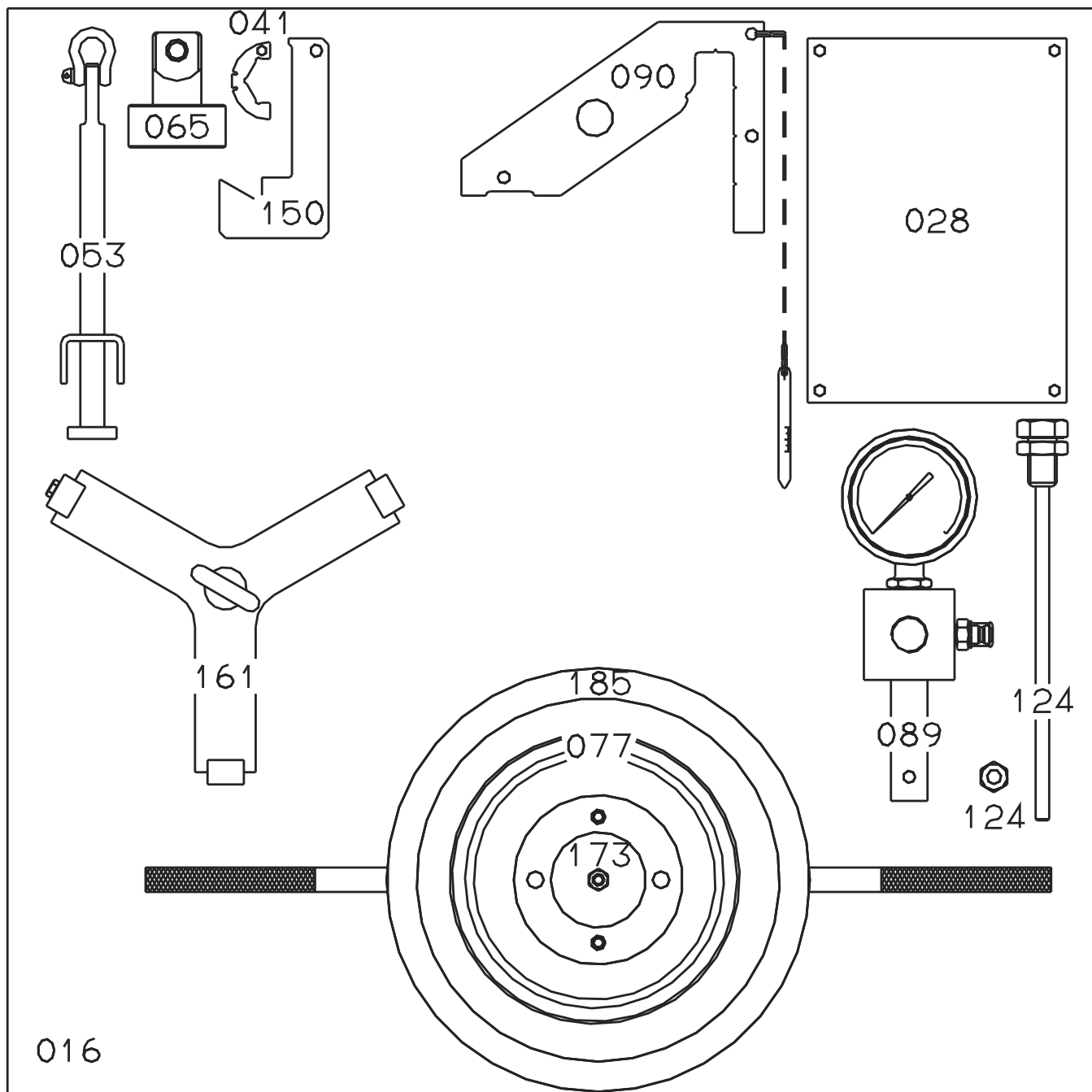


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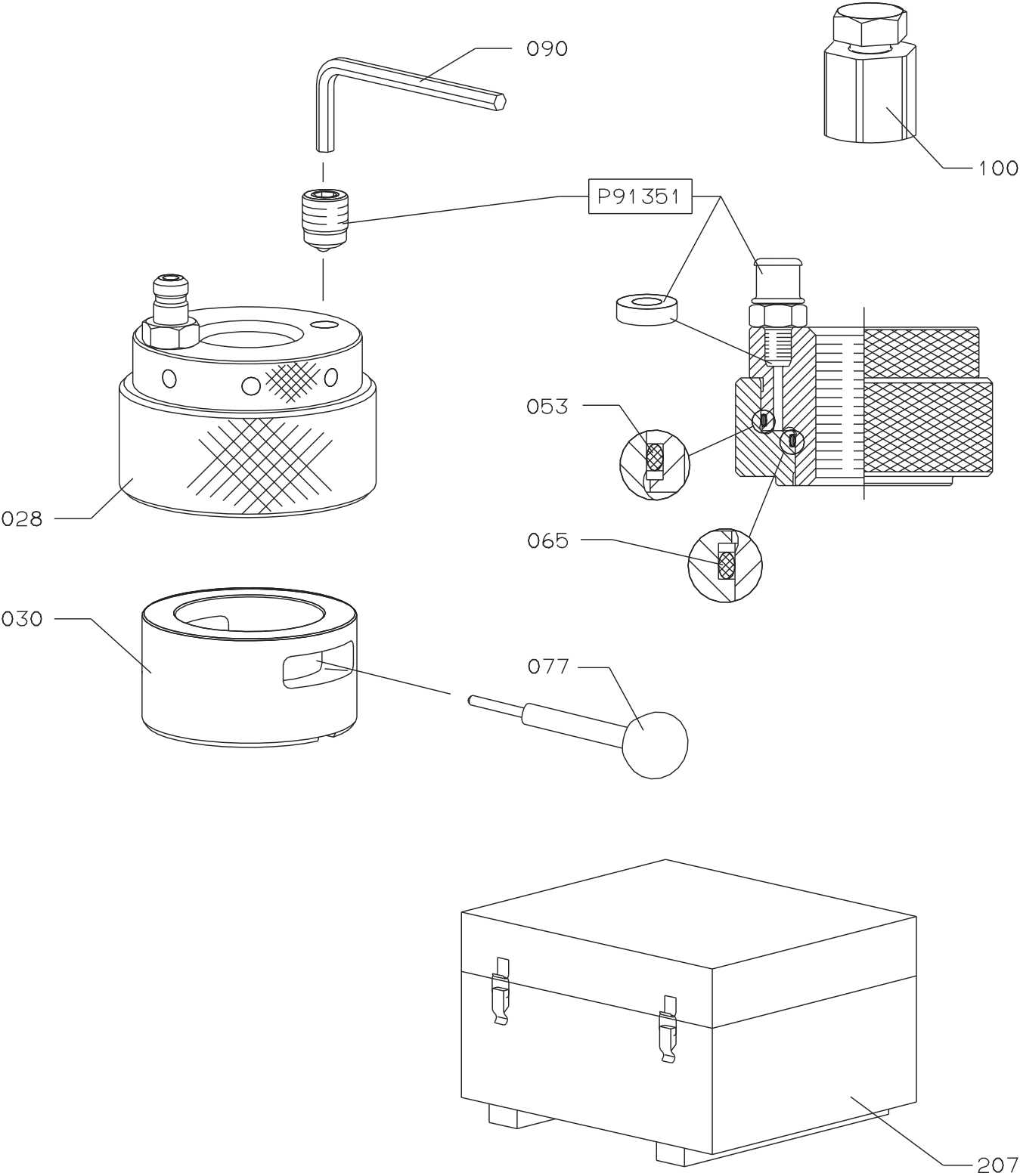
11.



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Item No.	Item Description		
016	Panel for tools		
028	Name plate		
041	Tightening template, actuator		
053	Lifting tool , roller guide		
065	Lifting tool, exhaust valve		
077	Conical ring, pneumatic piston		
089	Pressure tester, safety valve		
090	Gauge, exhaust valve spindle		
124	Tool, emergency opening of exhaust valve		
150	Gauge, exhaust valve bottom piece		
161	Lifting tool, exhaust valve spindle		
173	Lifting tool, exhaust valve roller guide		
185	Grinding ring, exhaust valve bottom piece		



Item No.	Item Description		
028	Jack-hydraulic, complete		
030	Support		
053	O-ring with back-up ring		
065	O-ring with back-up ring		
077	Tommy bar		
090	Spanner		
100	Stud setter		
207	Hydraulic toolset, complete		